VidyaLok: An AI-Powered Smart Library Ecosystem using NLP and Predictive Models for APSIT

Submitted in partial fulfilment of the requirements of the degree of

BACHELOR OF ENGINEERING

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A. P. SHAH INSTITUTE OF TECHNOLOGY, THANE

(2025-2026)

A. P. SHAH INSTITUTE OF TECHNOLOGY, THANE

CERTIFICATE

This is to certify that the project entitled "VidyaLok: An AI-Powered Smart Library Ecosystem using NLP and Predictive Models for APSIT" is a bonafide work of Krishna Mishra (22106131), Ayush Gondhali (22106124), Harsh Jain (22106117), Tanuj Kokamkar (22106088) submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of Bachelor of Engineering in Computer Science & Engineering (Artificial Intelligence & Machine Learning).

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Project Report Approval for B.E.

This project report entitled "VidyaLok: An AI-Powered Smart Library Ecosystem using NLP and Predictive Models for APSIT" by "Krishna Mishra, Ayush Gondhali, Harsh Jain and Tanuj Kokamkar" is approved for the degree of *Bachelor of Engineering* in *Computer Science & Engineering (Artificial Intelligence & Machine Learning), 2025-26.*

Examiner Name	Signature
1._____	_____
2._____	_____

Date:

Place:

Declaration

We declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

Libraries are an essential part of academic institutions, but conventional systems often remain limited to cataloging, circulation, and basic searches, offering little support for personalization or interactive learning. This gap reduces student engagement and makes it harder for administrators to manage resources effectively. To address these challenges, VidyaLok has been designed as an AI-augmented library ecosystem that transforms the traditional library into a smart, student-centric environment. The system introduces features such as personalized recommendations, a conversational assistant for query resolution, automated chapter summarization, sentiment-based feedback analysis, occupancy monitoring with predictive insights, and instant notification alerts during emergencies. Together, these features enhance accessibility, efficiency, and student satisfaction. By integrating intelligent automation and predictive capabilities, VidyaLok not only modernizes daily library operations at APSIT but also contributes to the larger vision of smart campus infrastructure and future educational innovation.

Keywords: AI-augmented library, personalized learning, chatbot, summarization, sentiment analysis, occupancy forecasting, smart campus.

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ABBREVATION

AI	*Artificial Intelligence*
ML	*Machine Learning*
NLP	*Natural Language Processing*
IoT	*Internet of Things*
QR	*Quick Response (code)*
CF	*Collaborative Filtering*
LSTM	*Long Short-Term Memory*
APSIT	*A.P. Shah Institute Of Technology*

Chapter 1
Introduction

Libraries have always been central to education, serving as repositories of knowledge and resources for students and researchers. With the rapid growth of academic content and digital learning resources, conventional library systems are struggling to keep up with evolving user needs. Most existing systems are limited to catalog management, keyword-based search, and manual record-keeping, which do not provide the intelligence or personalization today's learners expect. In the era of smart campuses and digital education, students need quick access to relevant materials, interactive assistance, and intelligent tools that support effective learning.

Indian academic institutions, in particular, face challenges of large student populations, limited library staff, and outdated systems. Students often spend excessive time locating relevant books, waiting in queues, or manually checking availability. Administrators lack real-time analytics for space utilization or feedback analysis, and libraries are unable to provide modern features such as summarization, or automated emergency alerts. To overcome these limitations, there is a strong need for an AI-powered ecosystem that makes libraries more intelligent, secure, and student-centric.

Motivation

Today's students are used to personalized digital services, yet most library systems remain static, offering uniform experiences that reduce engagement and limit the use of resources. Conventional systems lack features like tailored book suggestions, concise summaries, or auto-generated quizzes that can support exam preparation and active learning. Similarly, librarians struggle without intelligent tools for occupancy monitoring, resource tracking, feedback analysis, and emergency communication. VidyaLok is motivated by these gaps and aims to embed AI/ML modules that bring personalization, interactivity, and automation into the library. By transforming it into a student-centric ecosystem, the project not only enhances daily operations at APSIT but also establishes a model for smart campus infrastructure and future EdTech innovation.

Role of AI/ML in Libraries

Artificial Intelligence and Machine Learning (AI/ML) are transforming academic libraries from static repositories into intelligent, student-focused learning hubs. With tools like recommendation engines, semantic search, and chatbots, libraries can provide personalized access to resources tailored to each student's academic needs. Machine learning also supports predictive analytics for occupancy, space optimization, and book demand forecasting, improving efficiency. Natural language processing adds further value by enabling summarization and feedback analysis. By integrating these technologies, libraries become a vital part of smart campus infrastructure, enhancing learning, administration, and overall campus safety.

The Benefits of VidyaLok

VidyaLok is proposed as an AI-augmented library management and engagement ecosystem for APSIT. It integrates advanced AI/ML modules to transform the library into an intelligent educational assistant. Key features include:

- Personalized Recommendation Engine to suggest books based on branch, semester.
- AI Chatbot for natural language interaction and query resolution.
- Transformer-based Summarizer for generating concise chapter-wise digests.
- Sentiment Analysis to evaluate student feedback automatically.
- Occupancy Monitoring & Forecasting with QR-based entry/exit logs and predictive analytics.
- Semantic Search to understand and respond to natural language queries.

In conclusion, VidyaLok reimagines the library as an AI-driven ecosystem that combines personalization, automation, and predictive intelligence to bridge existing gaps. It marks a step toward a smart campus at APSIT while serving as a prototype for future EdTech innovation.

Chapter 2

Literature Survey

Year	Title	Domain	Algorithm	Dataset	Gap
2025	AI-Powered Library Management System	Library Systems (Automation/ NLP)	NLP search, content-based and recommendations, IoT/RFID integration	N/A (theoretical framework)	Traditional library processes are manual and slow; need AI to automate tasks, personalize recommendations, and improve efficiency
2025	Sentiment Analysis in Learning Management Systems: Understanding Student Feedback at Scale	EdTech (NLP/ Sentiment	Word embeddings + LSTM with attention; logistic regression baseline	RateMy-Professor student feedback	Sentiment analysis of student feedback in LMS is under-explored; existing systems lack tools to interpret qualitative feedback and bridge communication gaps
2024	Lib-Bot: A Smart Librarian Chatbot Assistan	Library Services (Chatbot/NLP)	BERT for intent classification; speech-to-text; Naïve Bayes/SVM fallback	Library FAQ and query logs	Users often cannot reach librarians); need an AI chatbot to handle common inquiries via

					text/voice.
2024	QR Code Integrated ML Analytics Library System	Smart Library (IoT/Smart Campus)	Unique QR-code mapping + ML rule-based system	N/A (conceptual)	Need of touchless library solution. Proposed system assigns unique QR IDs to users, enabling touch-free check in/out and simplified inventory with minimal overhead
2022	A Hybrid Book Recommendation System for University Library	Library/ Recommender	Hybrid: content based + collaborative filtering	University library catalog + course syllabus data	Growing library collections make manual search slow and unsatisfying; existing systems lack recommendation features. This hybrid model leverages course syllabi to address cold-start and improve book discovery for users
2021	Library Book Intelligent Recommendation System Based on Artificial Intelligence	Library/ Recommender	Hybrid CF (user based + item based clustering)	User book ratings/ evaluation	Traditional recommenders yield higher errors in sparse data. This AI-based system cleans and clusters users, applying hybrid collaborative

					filtering.

Chapter 3

Limitations of Existing Systems

Modern academic libraries, while partially digitized, still face significant gaps in functionality and user engagement. These gaps hinder students' ability to access information efficiently and limit administrators from making data-driven improvements. Some of the most critical limitations of existing systems are:

1) Lack of Personalization: Traditional systems treat all users alike, without considering branch, semester, or interests. This reduces engagement and often leads to underutilization of resources.
2) Basic Search Mechanism: Most systems rely on simple keyword matching and cannot understand semantic queries, making it difficult for students to discover relevant resources efficiently.
3) Manual Entry/Exit Tracking: Attendance is usually logged manually or via RFID, which is error-prone and cannot provide real-time occupancy analytics or usage forecasts.
4) Absence of Interactive Assistance: There is no chatbot or virtual assistant to resolve queries outside working hours, limiting accessibility and student convenience.
5) No Automated Summarization: Students must read lengthy chapters or research papers in full since existing systems do not offer AI-generated summaries for quick learning.
6) Unanalyzed Feedback: Feedback is collected but rarely processed with AI tools, leaving common issues unnoticed and delaying meaningful improvements.

In short, while conventional library management systems serve basic purposes like cataloging and circulation, they lack intelligence, automation, and adaptability. These limitations clearly justify the need for an AI-augmented solution like VidyaLok, which addresses these shortcomings and transforms libraries into smarter learning environment

Chapter 4

Problem Statement, Objectives and Scope

Problem Statement

Traditional library management systems are limited to basic cataloging, circulation, and keyword-based searches, lacking personalization, semantic understanding, and intelligent automation. Students face difficulties in finding relevant books quickly, accessing summarized content, or receiving 24/7 support, while administrators cannot monitor occupancy, analyze feedback, or broadcast urgent notifications effectively. These shortcomings reduce student engagement, create inefficiencies, and prevent the library from evolving into a smart, secure, and student-centric learning environment.

Objectives

1) Personalized Recommendations – Suggest books based on branch, semester, and student interests.
2) AI-Powered Chatbot – Provide round-the-clock query resolution through natural language interaction.
3) Automated Summarization – Generate concise chapter digests to speed up the learning process.
4) Sentiment Analysis – Analyze feedback automatically for quicker resolution of complaints.
5) Occupancy Forecasting – Monitor seat usage and predict peak hours to optimize resources.

Scope

1) Academic Libraries – Designed primarily for college/university libraries to enhance student engagement.
2) Multi-Module Integration – Combines recommendation, chatbot, summarizer, sentiment analysis, and forecasting into one system.
3) Scalable Framework – Can be extended to other campuses and educational domains.

4) Smart Campus Support – Acts as a foundation for building AI-driven smart campus infrastructure.
5) Future Research – Scope for publishing research in EdTech, NLP, and AI-based library systems

Chapter 5

Proposed System

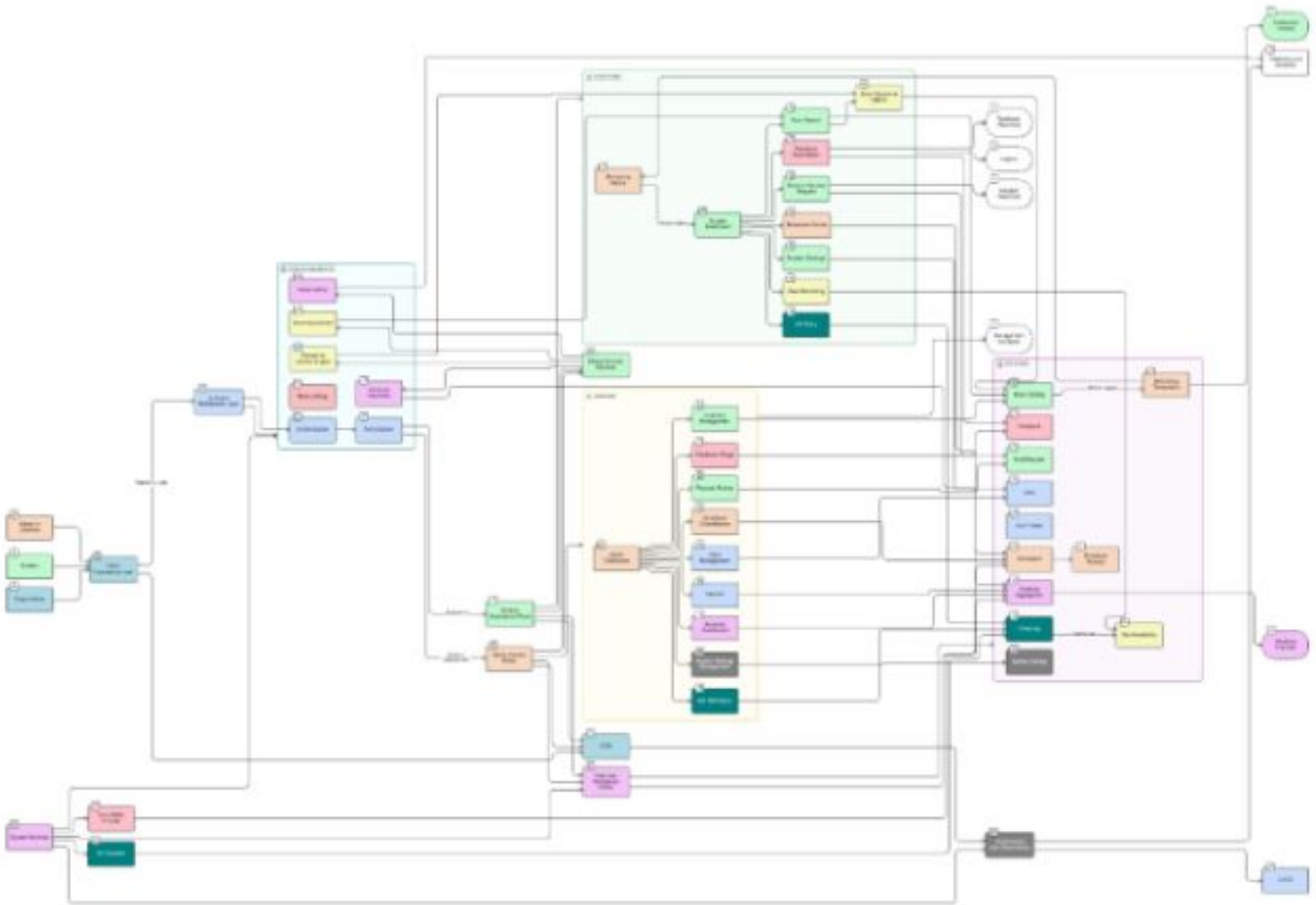


Figure 5.1 Architecture Diagram for VidyaLok

This diagram illustrates the complete workflow and interaction between the various components of the VidyaLok ecosystem. It captures how user requests, AI modules, backend services, and databases collaborate to provide a seamless, intelligent library experience. Key features include:

1) User Interface Layer: This layer includes students, librarians, and administrators who interact with the system through a web-based interface. Students can search for books, view recommendations, and access summaries, while librarians manage resources and monitor feedback through their dashboard.

2) Authentication and Access Control: Handles secure login and role-based authorization. Every request passes through the authentication module ensuring proper access rights for each user type before entering the core system.

3) Core Functional Modules: This section contains the main services of VidyaLok:

    - Recommendation Engine: Suggests relevant books based on user profiles and reading behavior using machine learning algorithms.

    - Chatbot Service: NLP-based virtual assistant that handles user queries and guides them to the right information.

    - Summarization Module: Uses NLP to generate short summaries of lengthy documents for faster understanding.

    - Book Search & Catalog Management: Enables searching, adding, or updating book records.

    - Feedback and Sentiment Analysis: Analyzes user feedback to understand satisfaction levels and provide insights to librarians.

    - Broadcast Center: Sends announcements, alerts, and important updates to users in real time.

    - Occupancy Prediction: Predicts library crowd levels using entry/exit data and assists in space management.

4) Middleware and Support Services: Includes modules such as Rate Limiting, Monitoring, and Automation that ensure smooth data flow, performance stability, and background process management.

5) Data Management Layer: Stores and retrieves all essential data. MongoDB Database is used to Maintains user data, book catalog, transactions, and feedback records.

6) Integration and External Services: Comprises QR/RFID-based entry tracking, notification service, and cloud deployment components. These services connect the library's physical environment with the digital system, ensuring real-time operations and scalability.

7) Admin Dashboard: Provides a centralized interface for librarians and administrators to oversee system performance, analyze user trends, and manage all core functionalities efficiently.

The architecture demonstrates a modular, multi-layered system that integrates frontend interaction, intelligent backend logic, and AI-based modules. Each layer works cohesively to deliver personalized recommendations, real-time support, and data-driven insights, thereby transforming APSIT's library into a smart, AI-augmented learning ecosystem.

Chapter 6

Experimental Setup

Software Requirements: -

Core Technologies

- **Node.js**: v18.18.0 or later
- **Next.js**: v15.4.5 (React framework)
- **React**: v19.1.0
- **TypeScript**: v5.9.2
- **MongoDB**: v6.0+
- **Prisma ORM**: v6.13.0

Authentication & Security

- **NextAuth.js**: v4.24.11
- **bcryptjs**: v3.0.2
- **Zod**: v3.25.76 (validation)

UI Framework

- **Tailwind CSS**: v4.1.11
- **Radix UI**: Component library
- **Lucide React**: v0.536.0 (icons)
- **Recharts**: v3.1.1 (charts)

Additional Libraries

- **qrcode**: v1.5.4
- **html2canvas**: v1.4.1
- **jsPDF**: v3.0.1
- **csv-parse**: v6.1.0

Hardware Requirements: -

Development Machine

- **Processor**: Intel Core i5 (8th Gen) or equivalent
- **RAM**: 8 GB minimum (16 GB recommended)
- **Storage**: 256 GB SSD
- **Display**: 1920×1080 resolution
- **Network**: 10 Mbps broadband

Production Server

- **vCPU**: 4 cores (2.5 GHz+)
- **RAM**: 8-16 GB
- **Storage**: 100 GB SSD with backup
- **Network**: 100 Mbps with CDN

Client Devices

- **Desktop/Laptop**: Intel Core i3, 4 GB RAM, Modern browser
- **Mobile**: Android 10+ or iOS 14+, 3 GB RAM

Operating Environment:

Development OS

- **Windows**: 10/11 (64-bit) with WSL2
- **macOS**: Monterey or later
- **Linux**: Ubuntu 20.04/22.04 LTS

Production Deployment

- **Primary**: Vercel (serverless)
- **Database**: MongoDB Atlas M10 cluster
- **Protocol**: HTTPS (TLS 1.3)

Browser Support

- Chrome/Edge: 100+
- Firefox: 95+
- Safari: 15+
- Mobile: iOS 14+, Android 10+

Development Environment:

IDE & Tools

- **Visual Studio Code**: v1.85+ with extensions:
    - ESLint
    - Prisma
    - Tailwind CSS IntelliSense
    - GitHub Copilot

Version Control

- **Git**: v2.40+
- **Repository**: GitHub (private)
- **CI/CD**: GitHub Actions

Package Management

- **npm**: v9.8+
- **Scripts**:
    - npm run dev (port 3000)
    - npm run build
    - npm run lint

Database Tools

- **MongoDB Compass**: v1.40+
- **Prisma Studio**: Built-in visual editor

Machine Learning & NLP Libraries:

- **TensorFlow.js: v4.x (client-side inference)**

- **Sentence-BERT: Semantic book search**
- **Hugging Face Transformers: Chatbot NLU**
- **scikit-learn: Collaborative filtering**

Chapter 7

Project Planning

Gantt Chart:

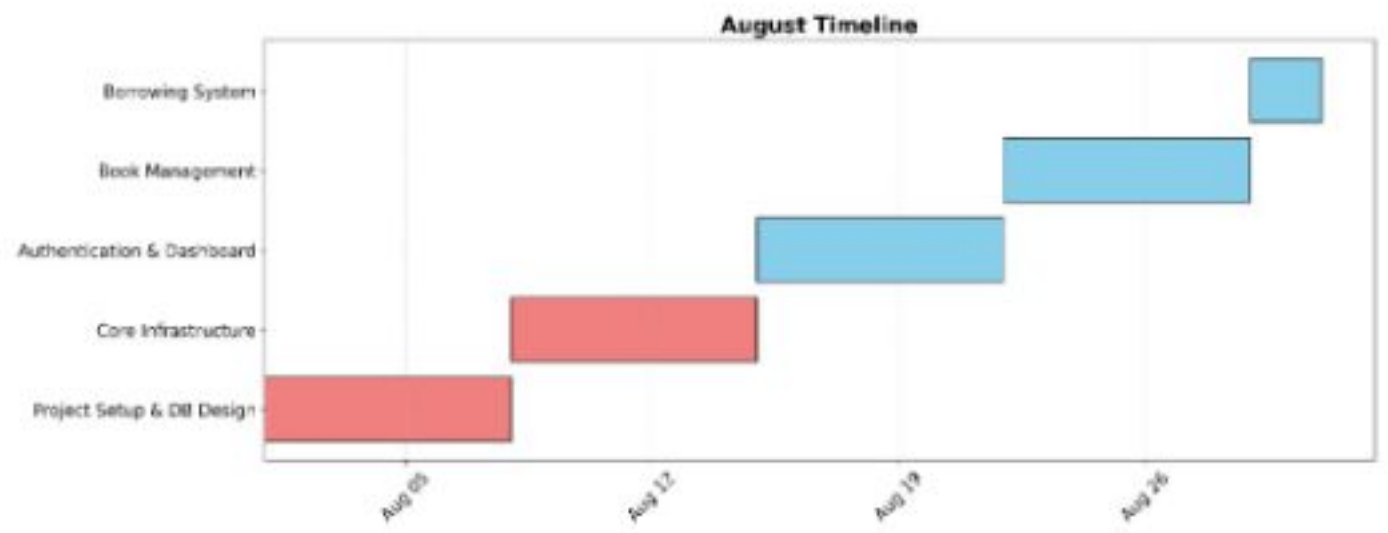


Figure 7.1 August (Gantt Chart)

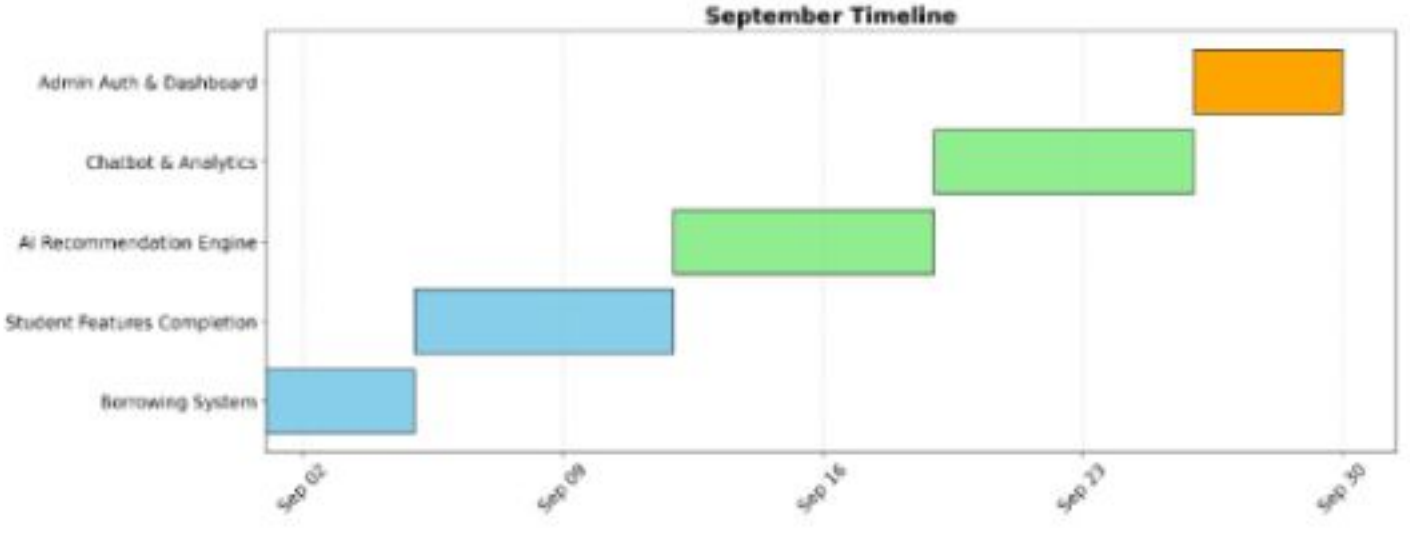


Figure 7.2 September (Gantt Chart)

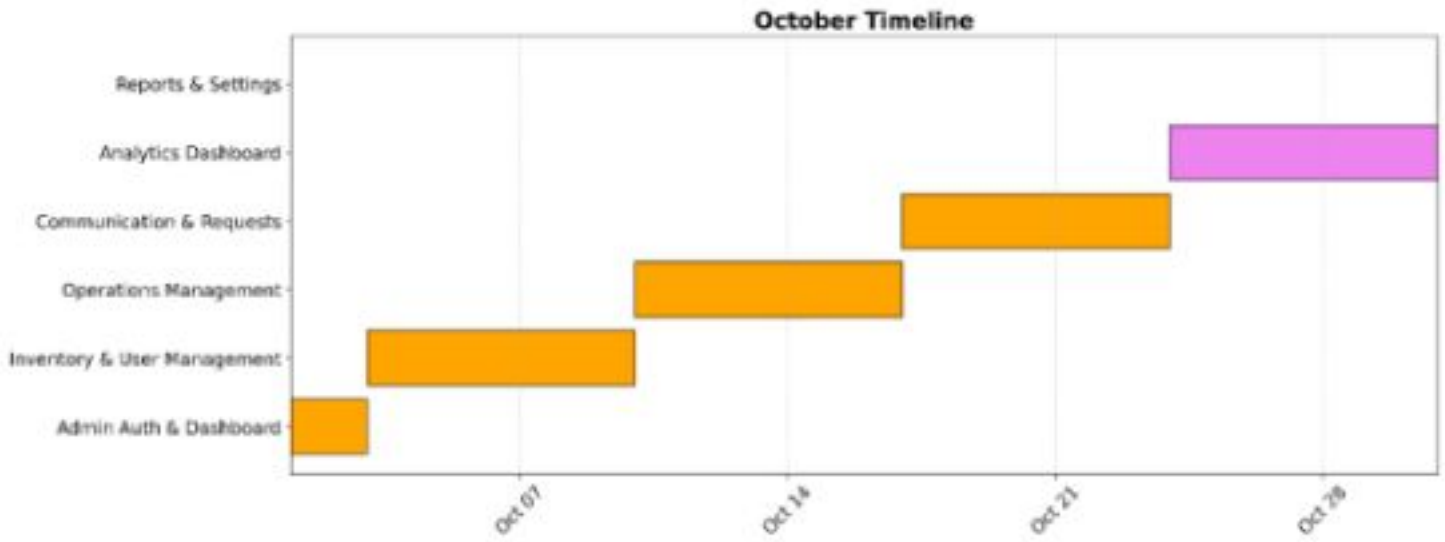


Figure 7.3 October & November (Gantt Chart)

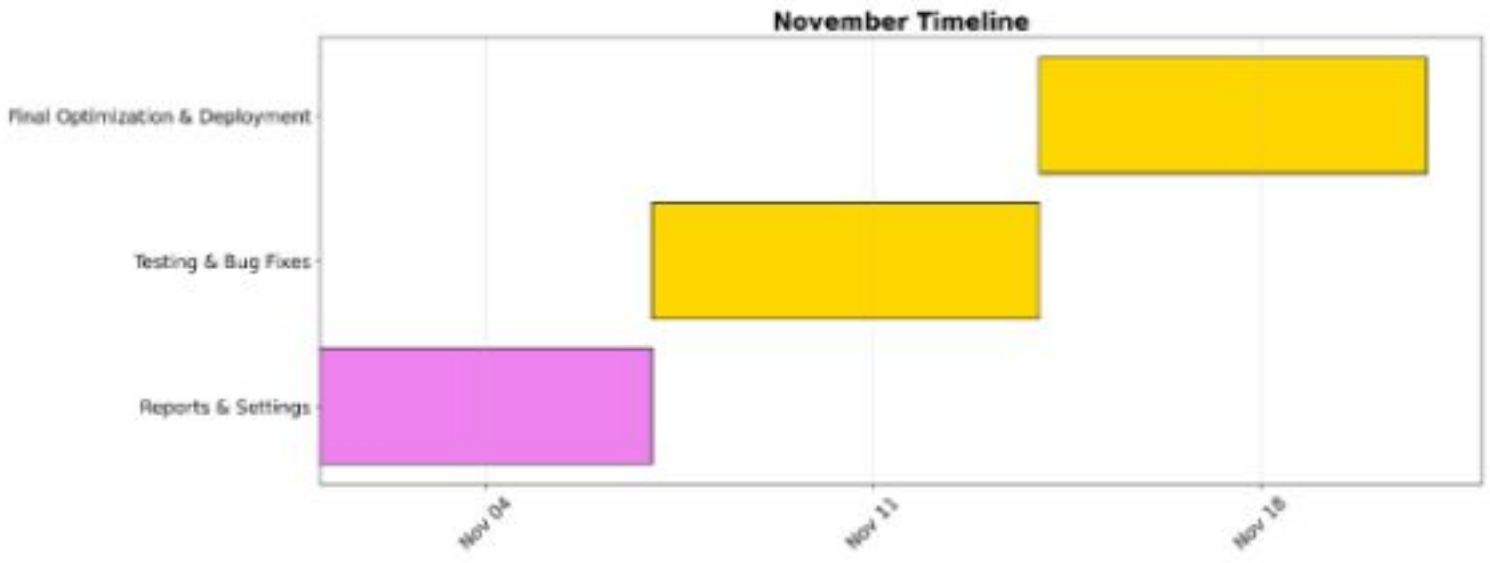
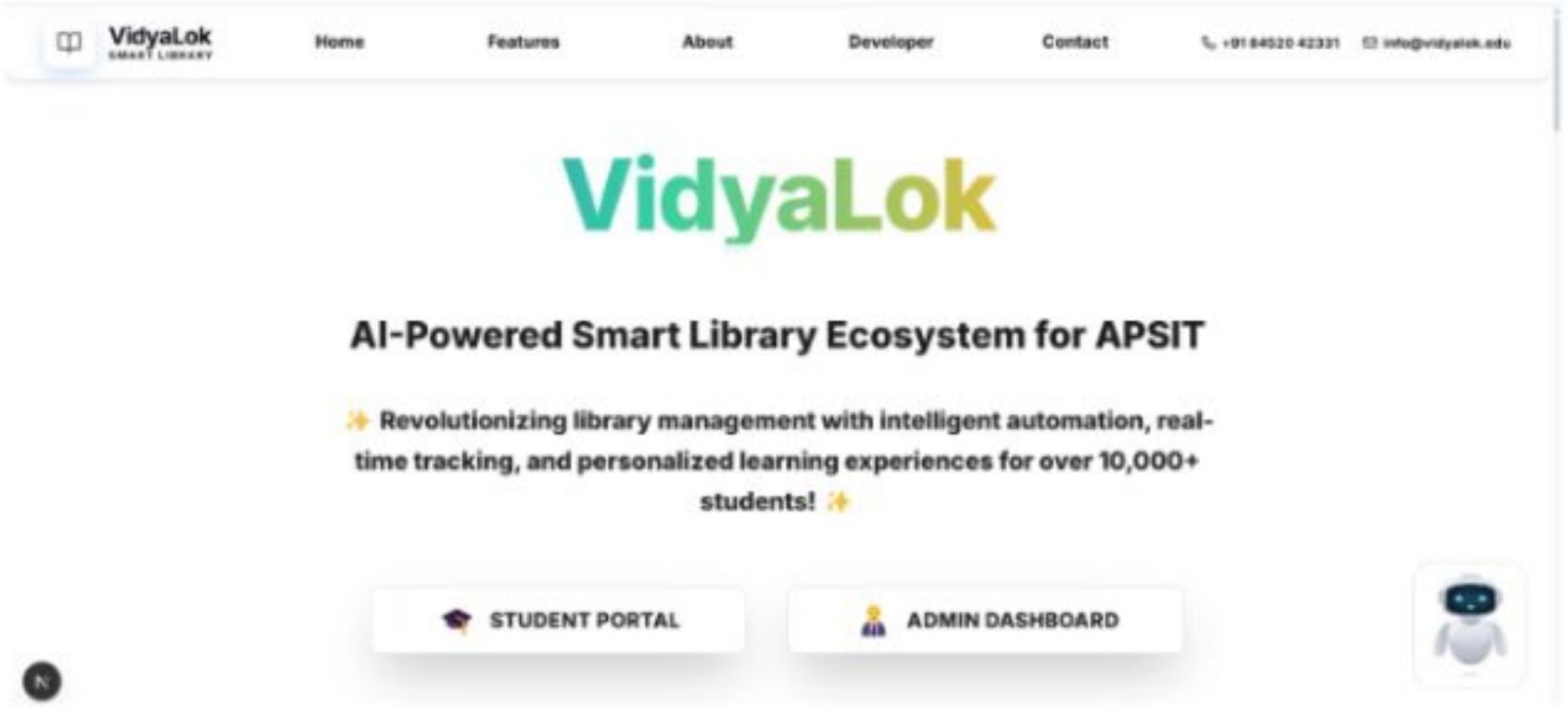


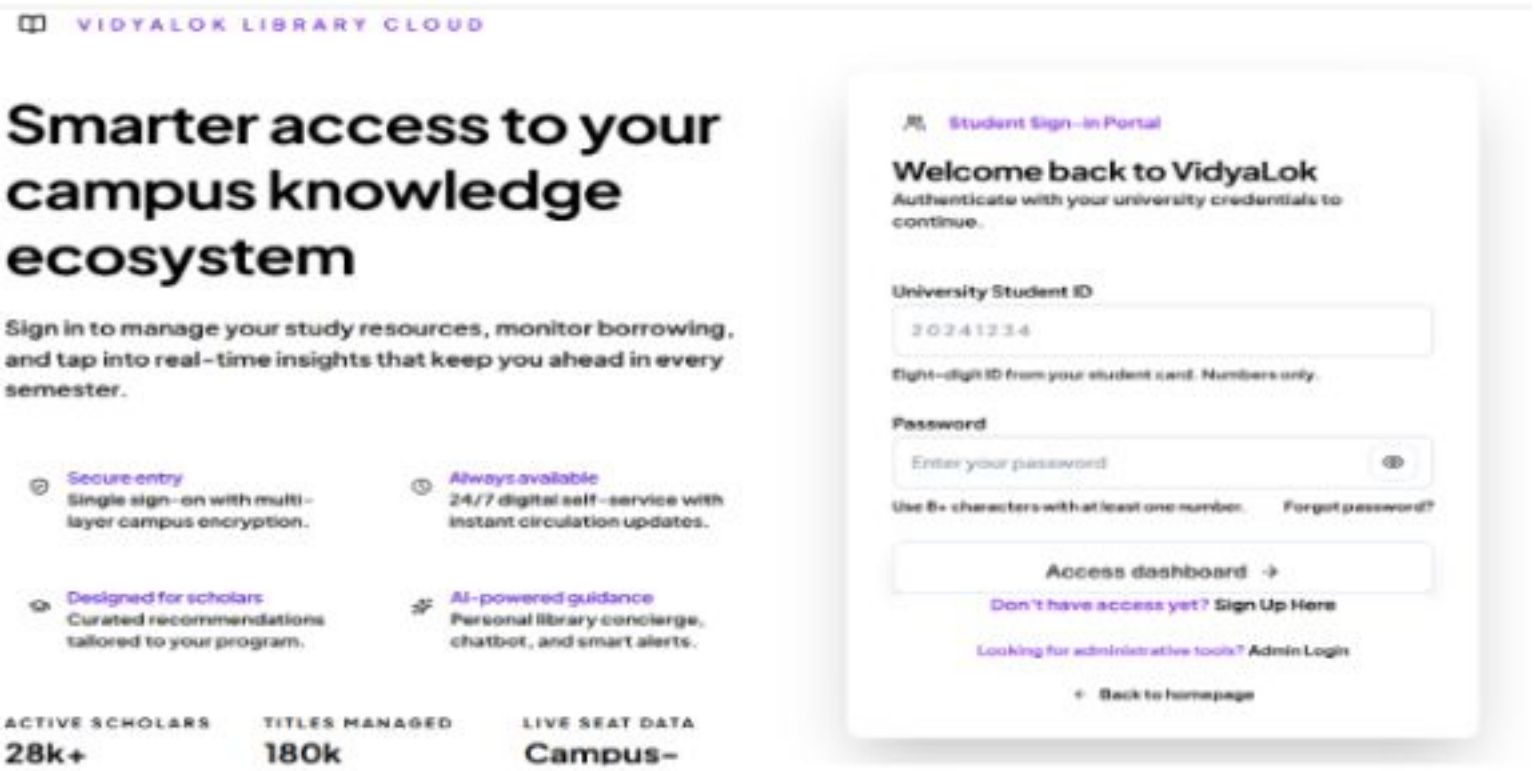
Figure 7.4 November (Gantt Chart)

Chapter 8

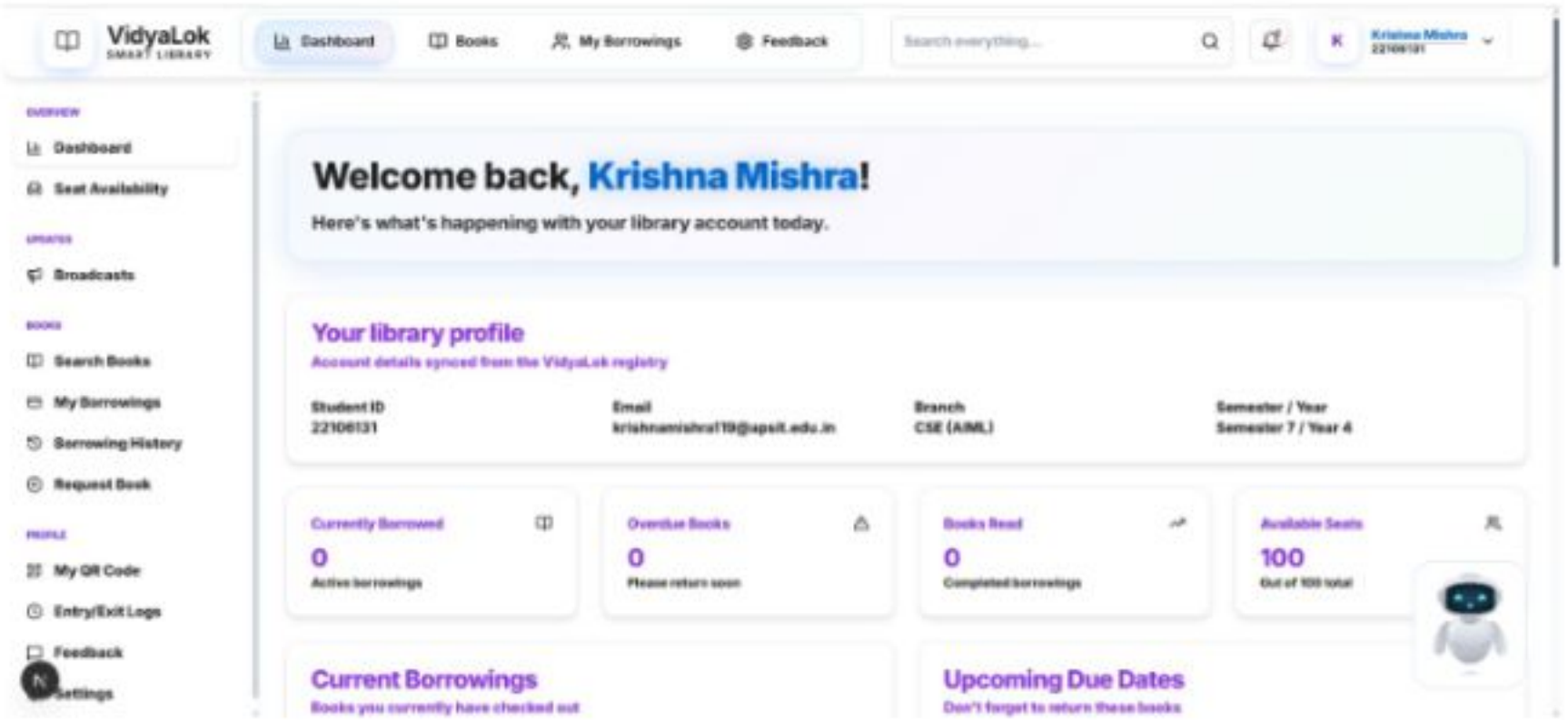
Expected Outcome



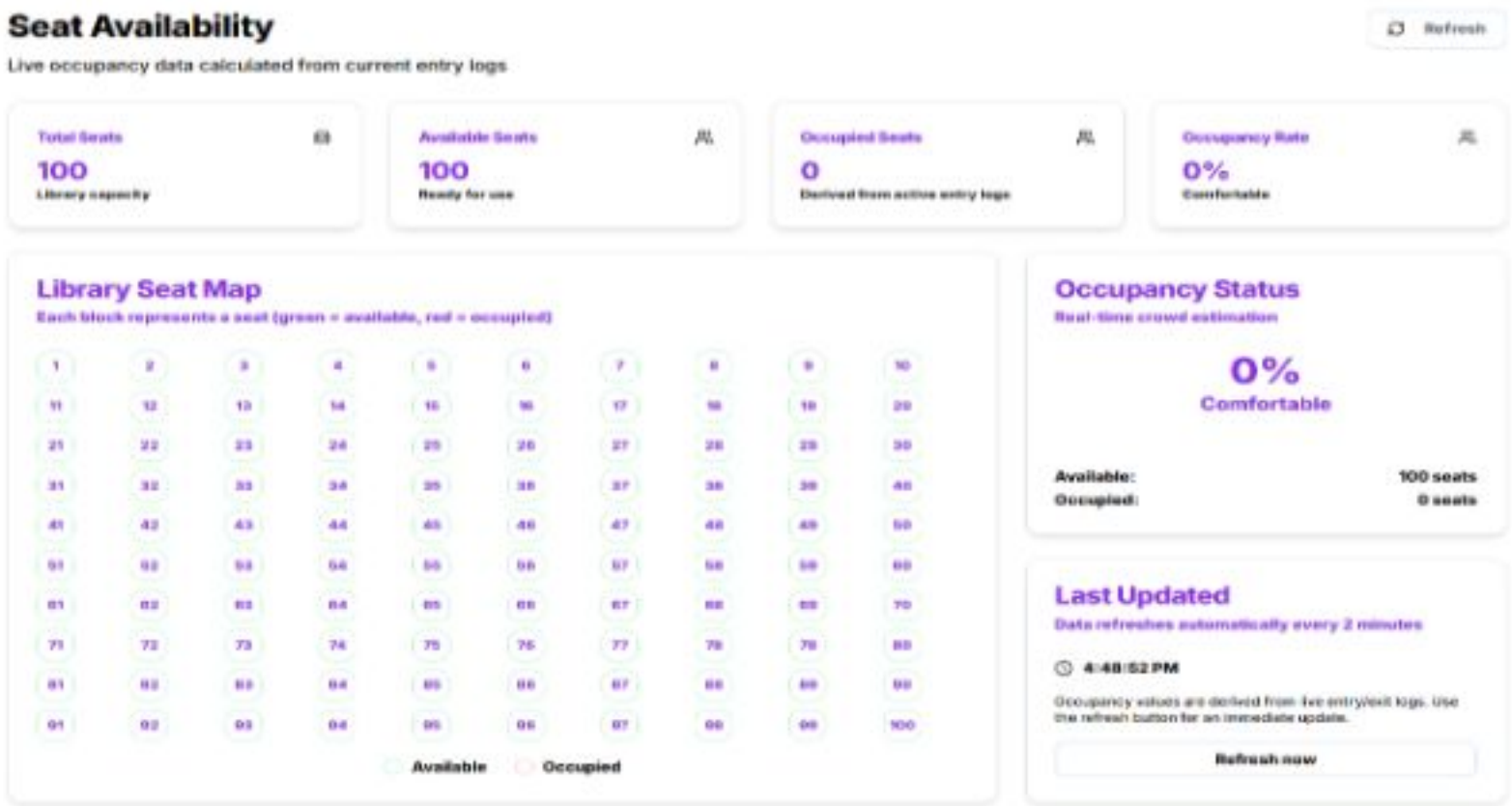
8.1 Home Page of VidyaLok



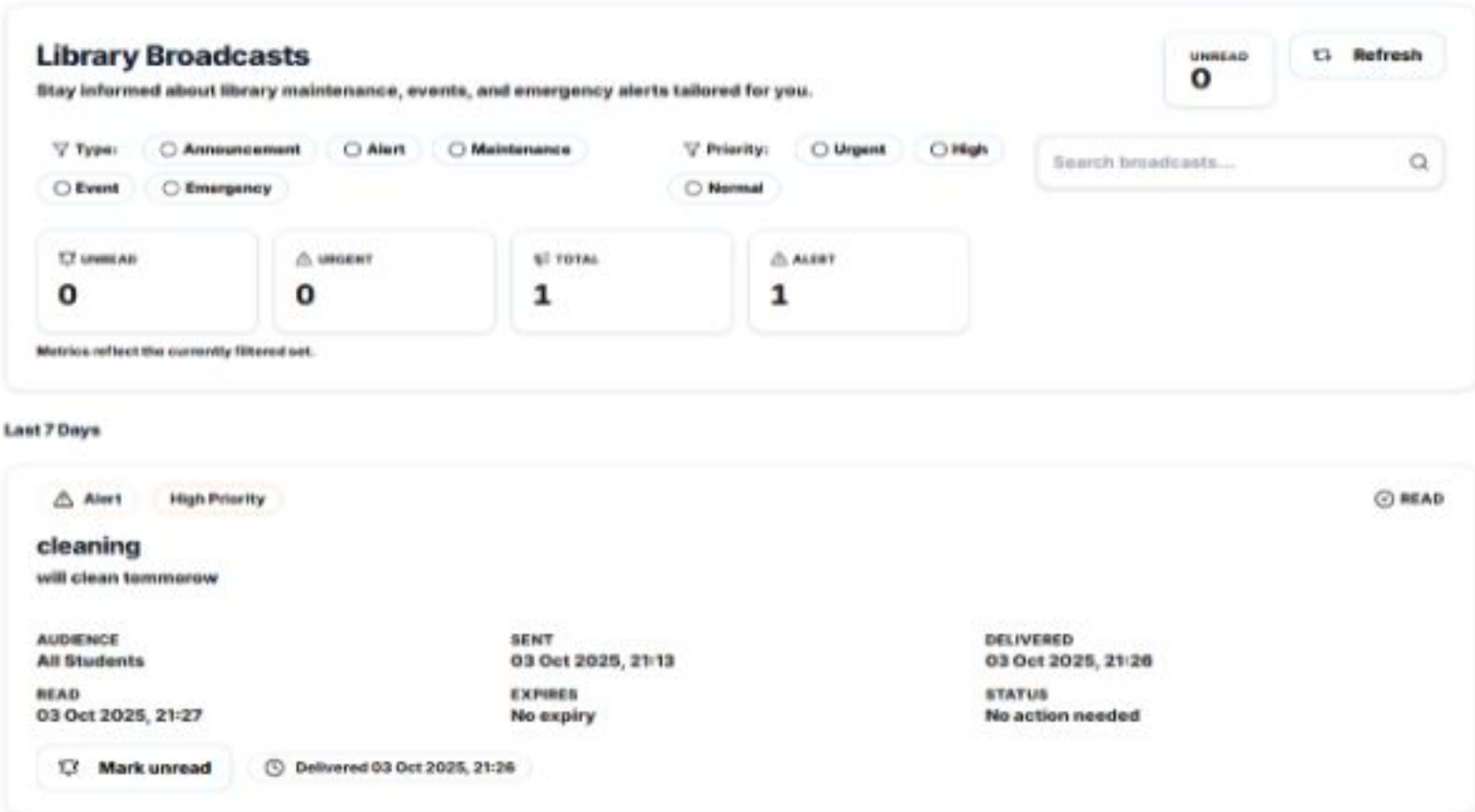
8.2 Student Sign-in Portal



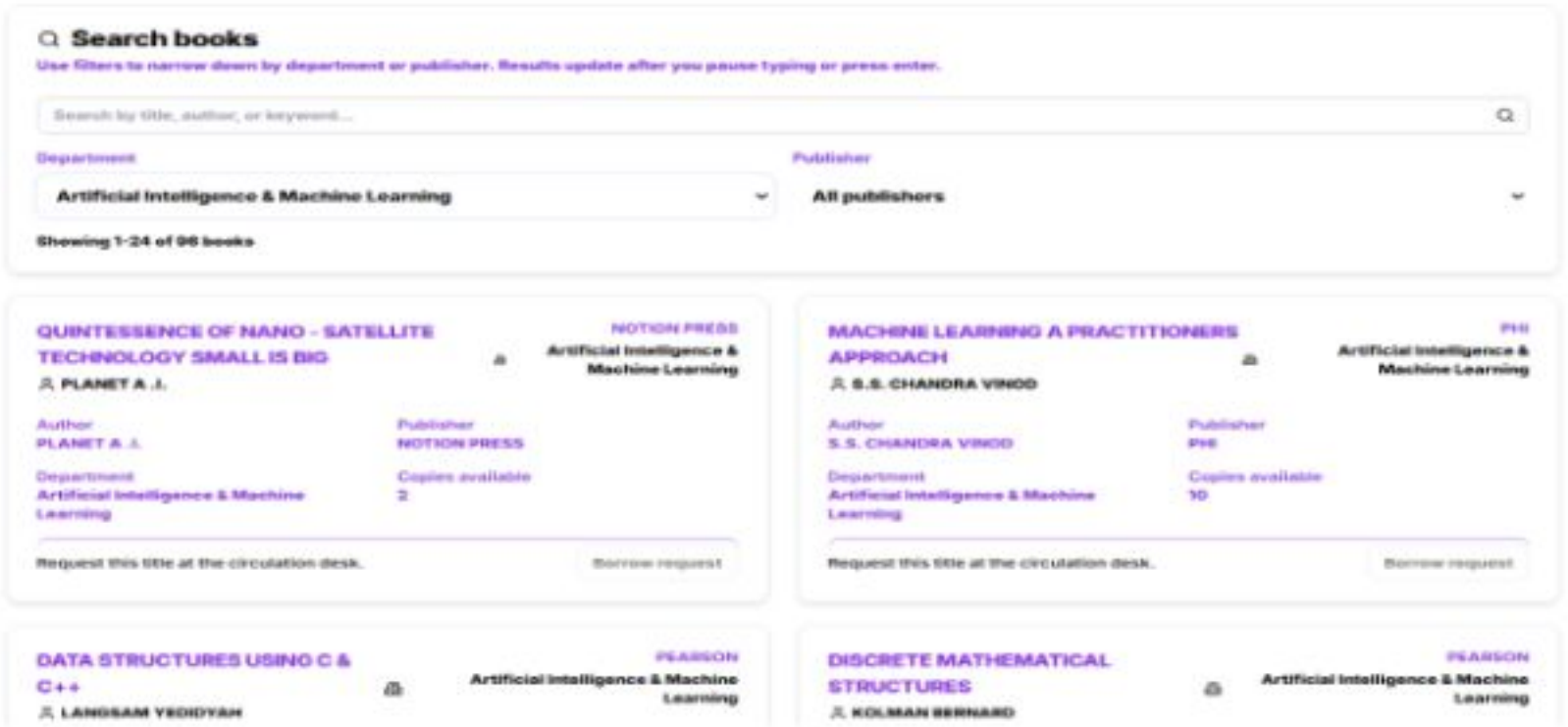
8.3 Student Dashboard



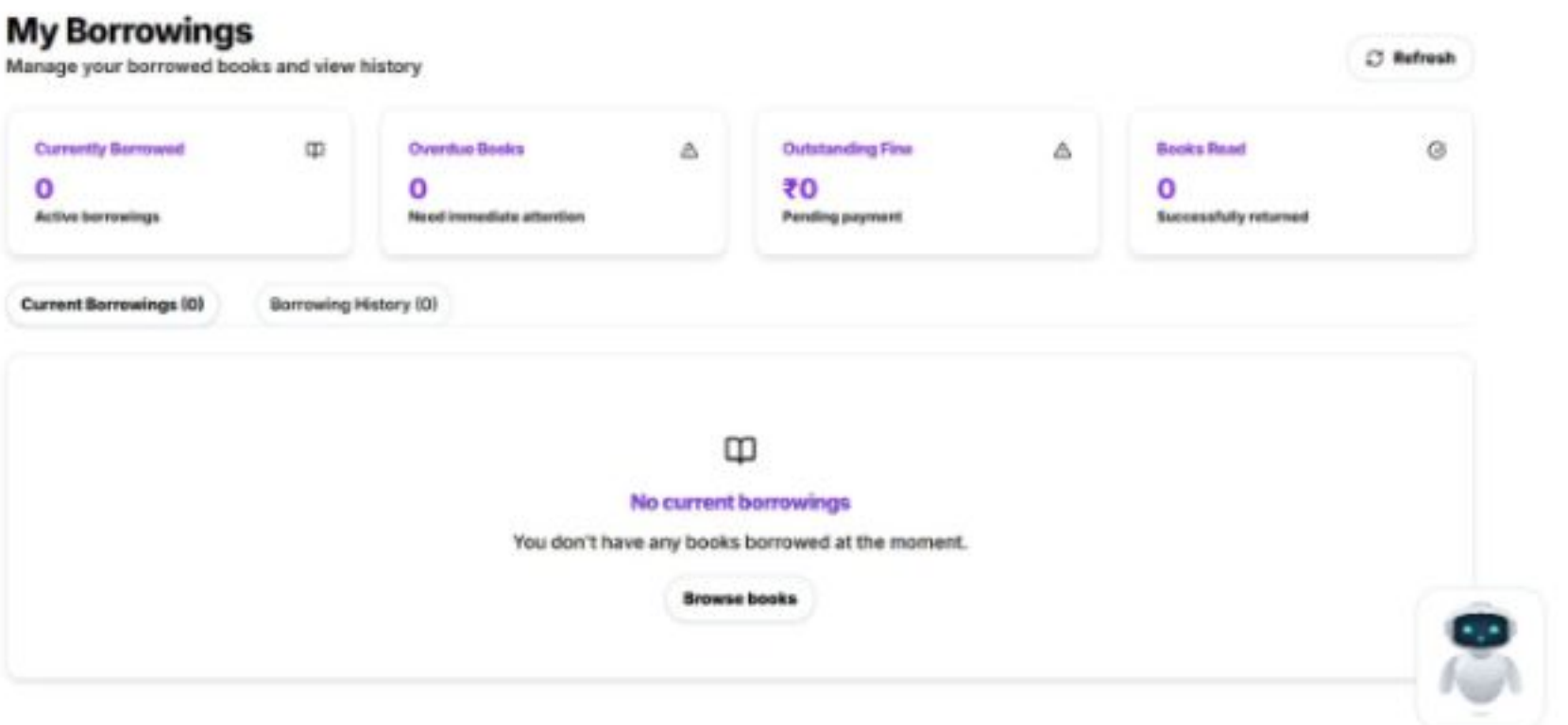
8.4 Seat Availability



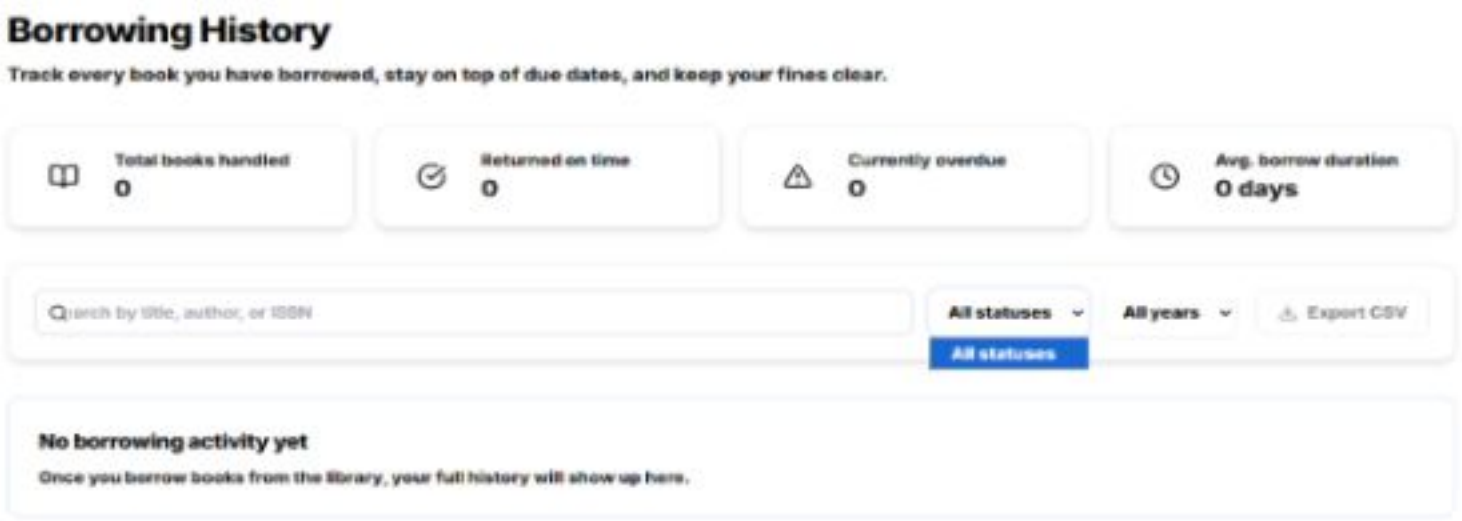
8.5 Library Broadcasts



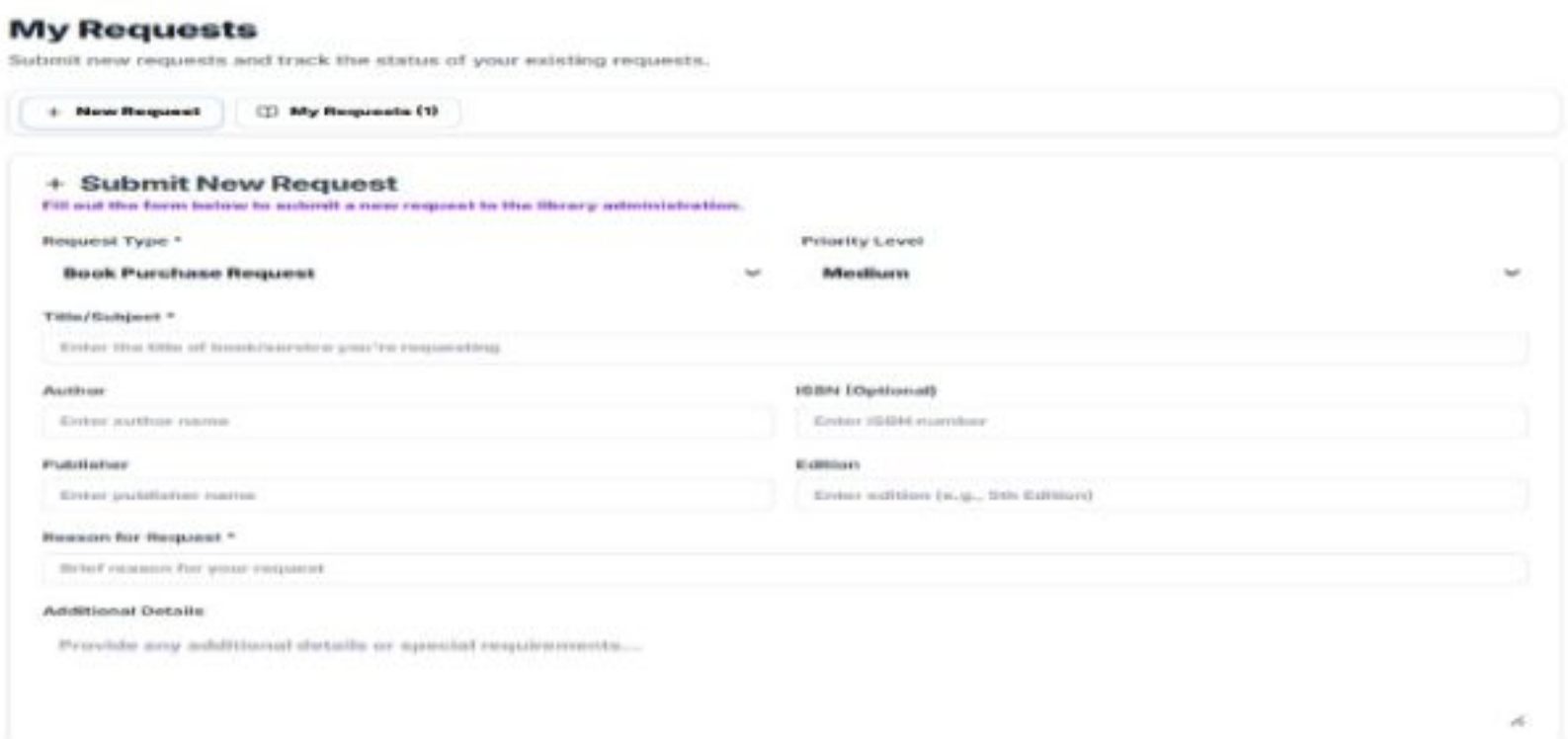
8.6 Search Book Page



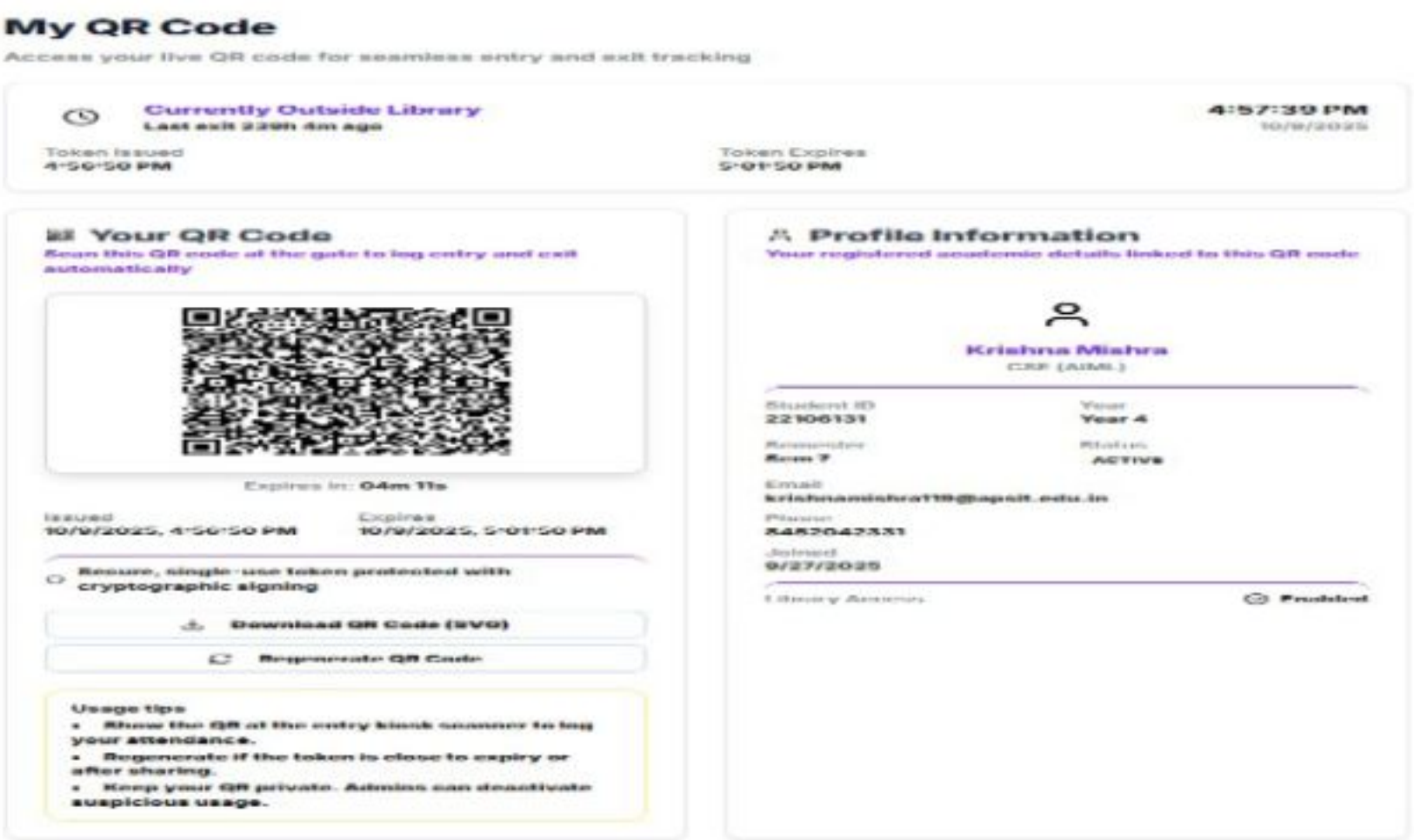
8.7 Student Borrowing



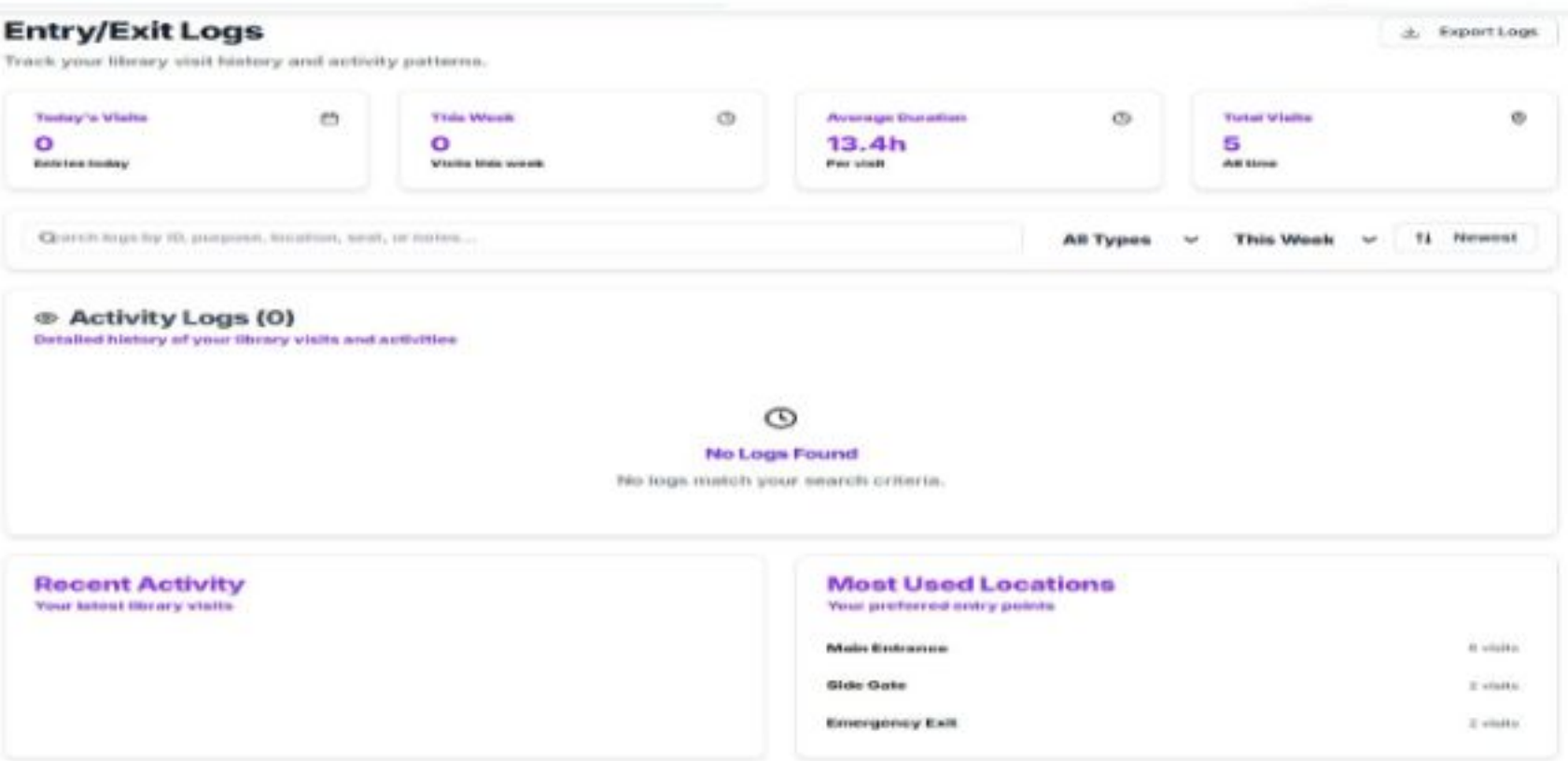
8.8 Borrowing History Page



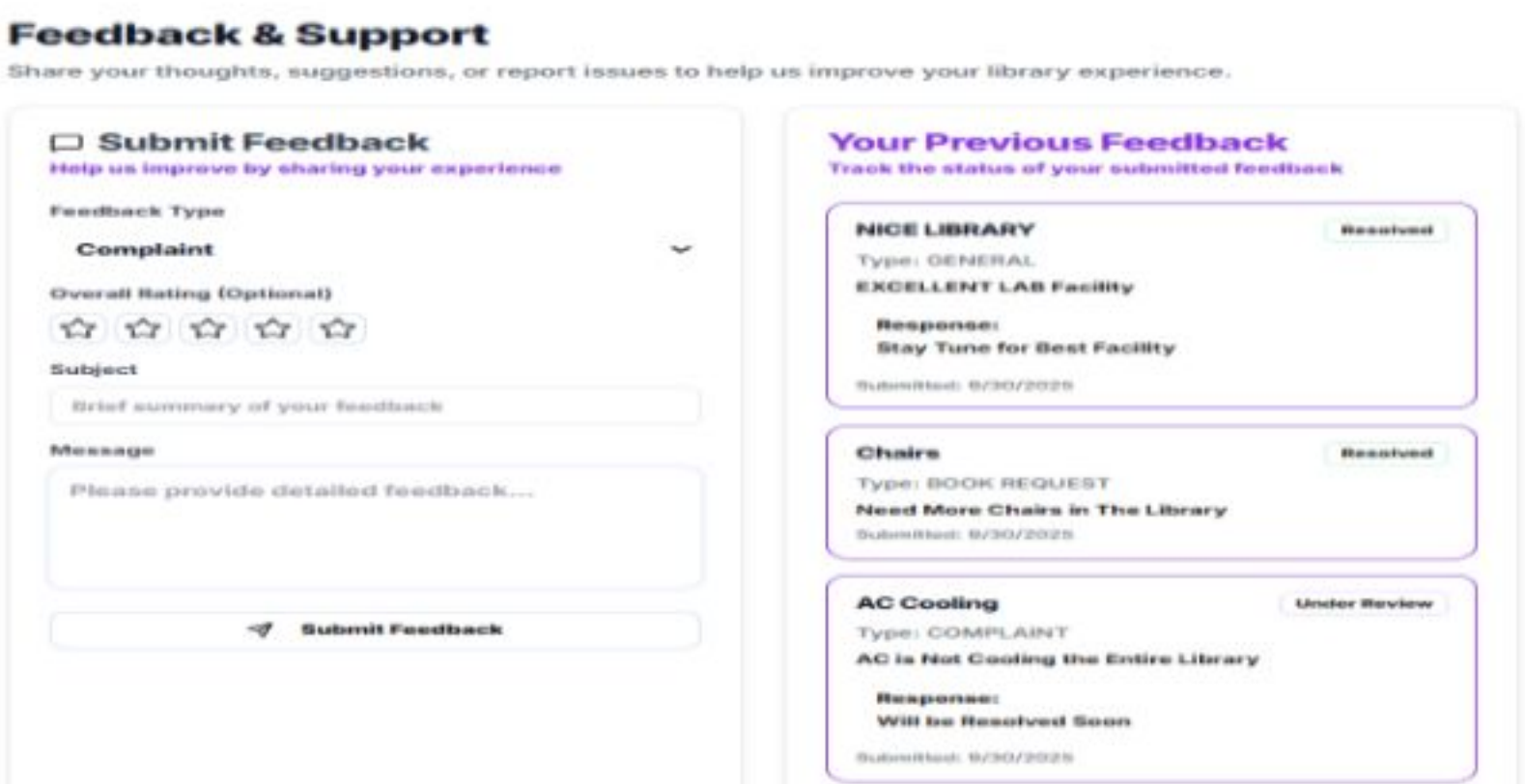
8.9 Book Request Page



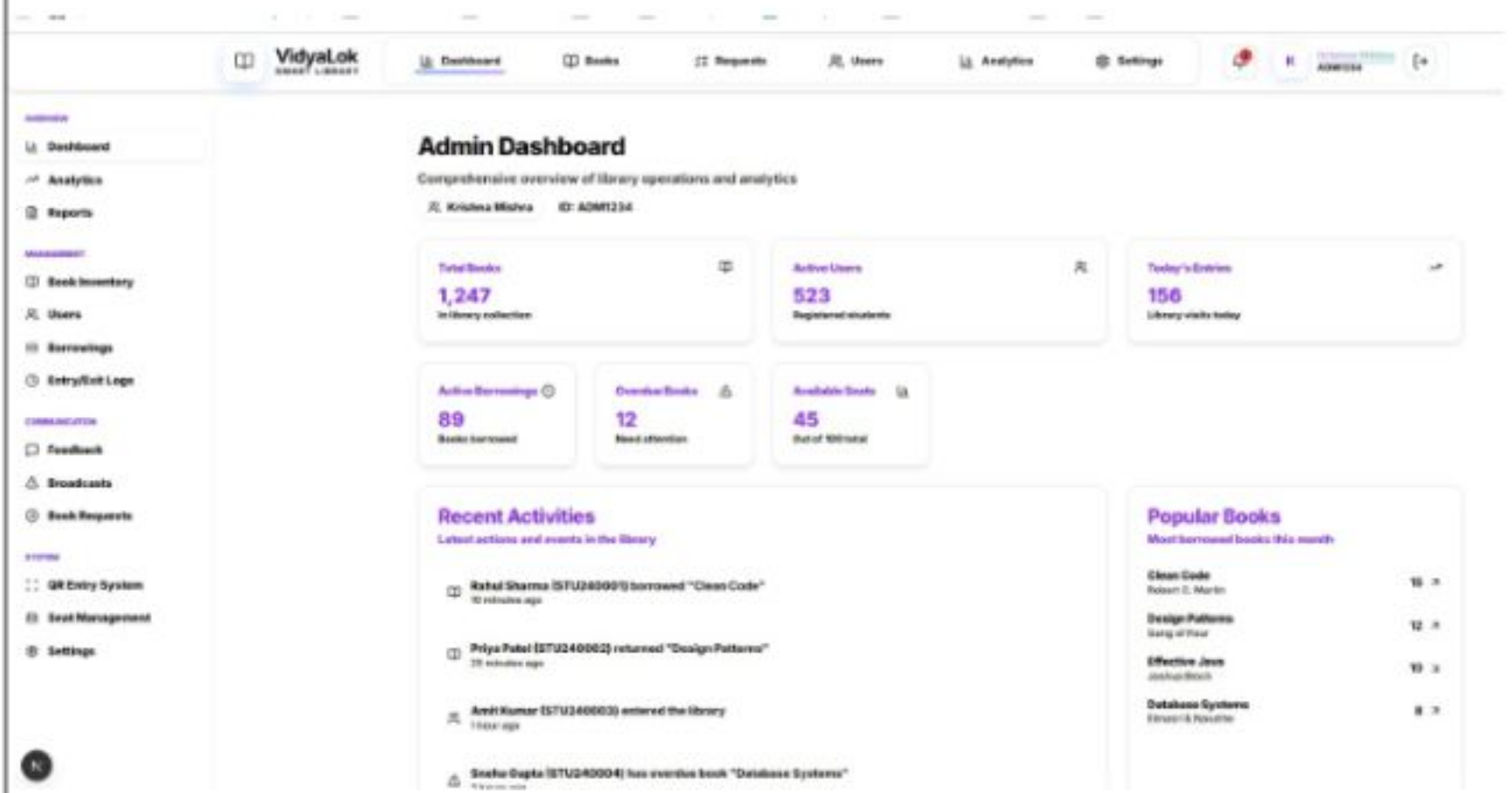
8.10 Student QR Code for Entry and Exit



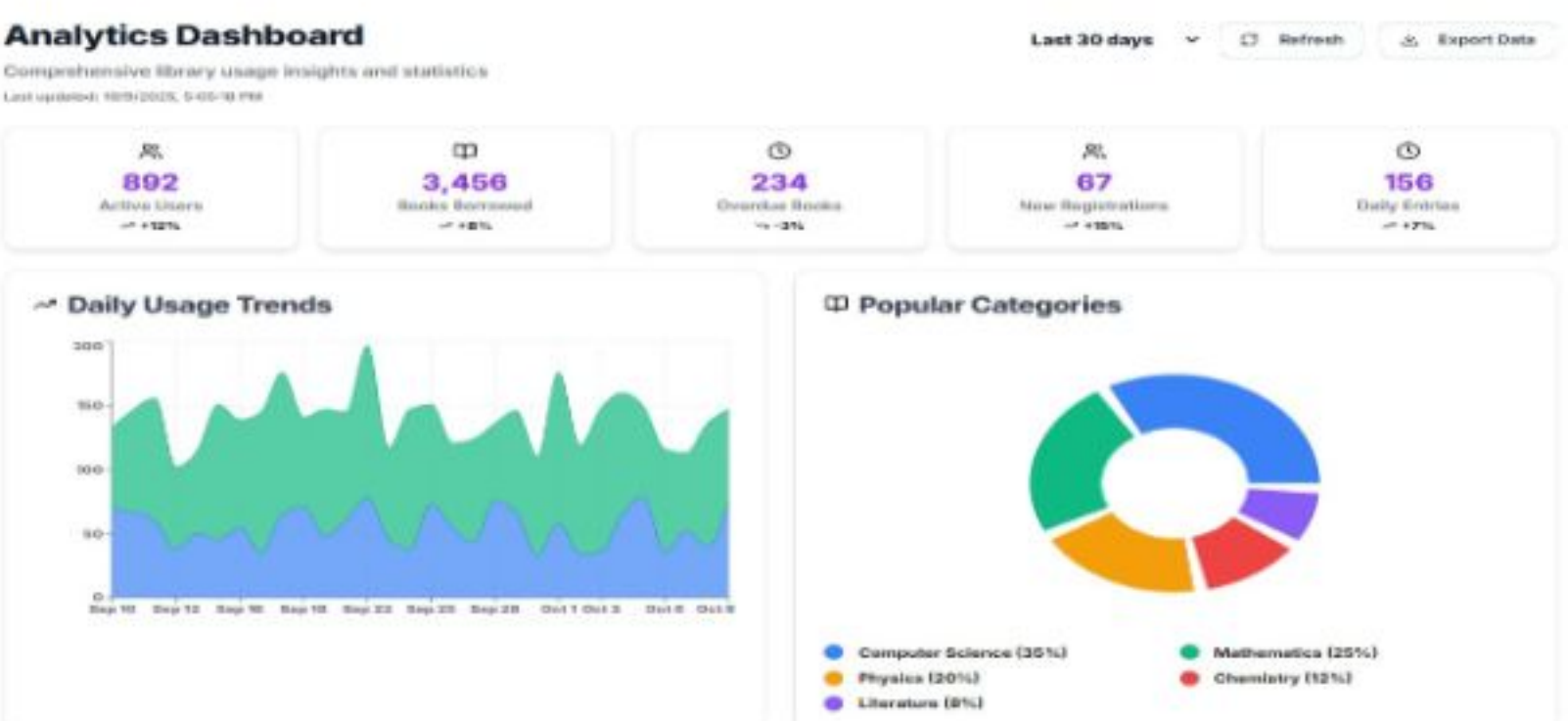
8.11 Student Entry/Exit Logs



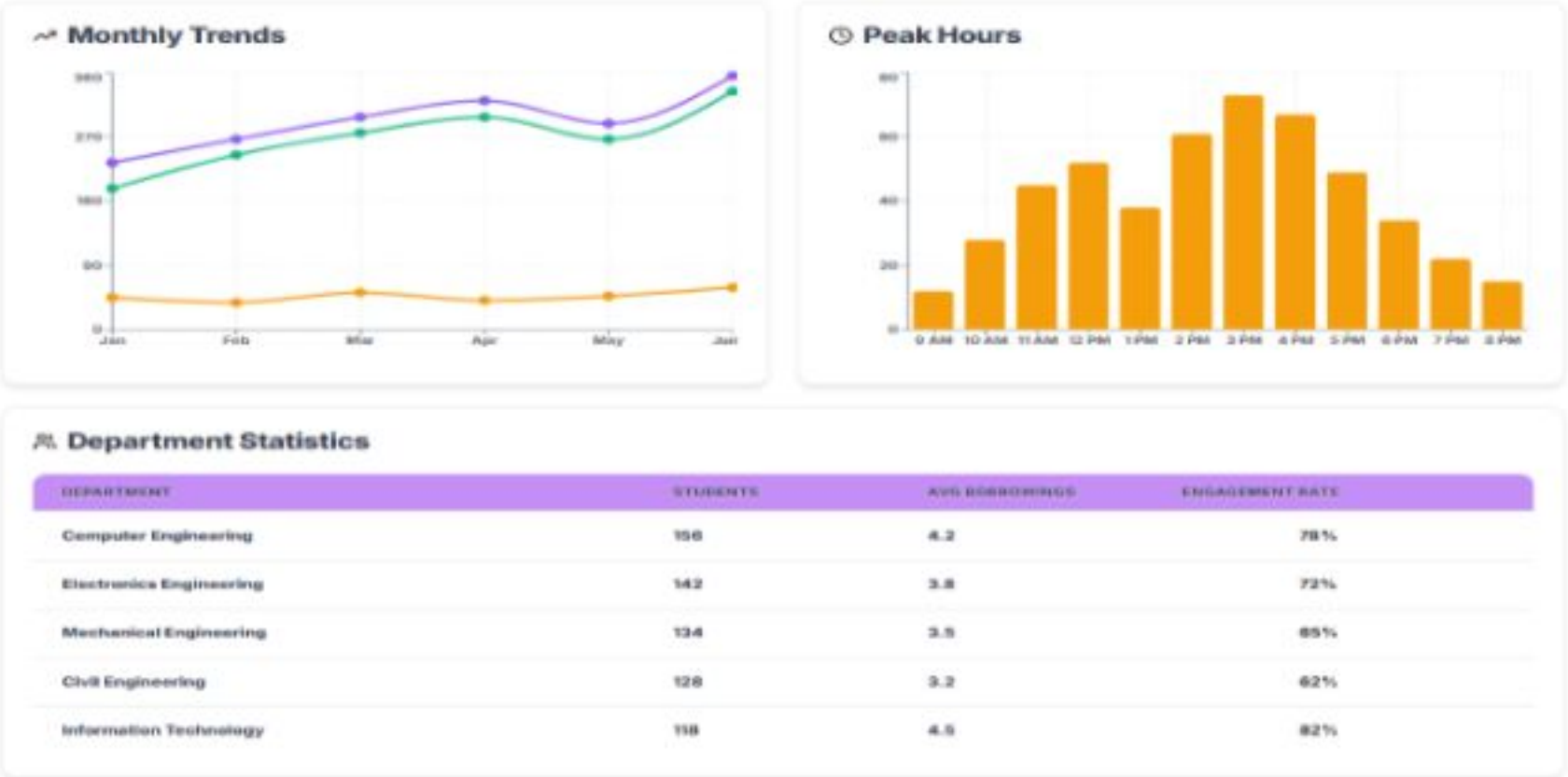
8.12 Student Feedback and Support page



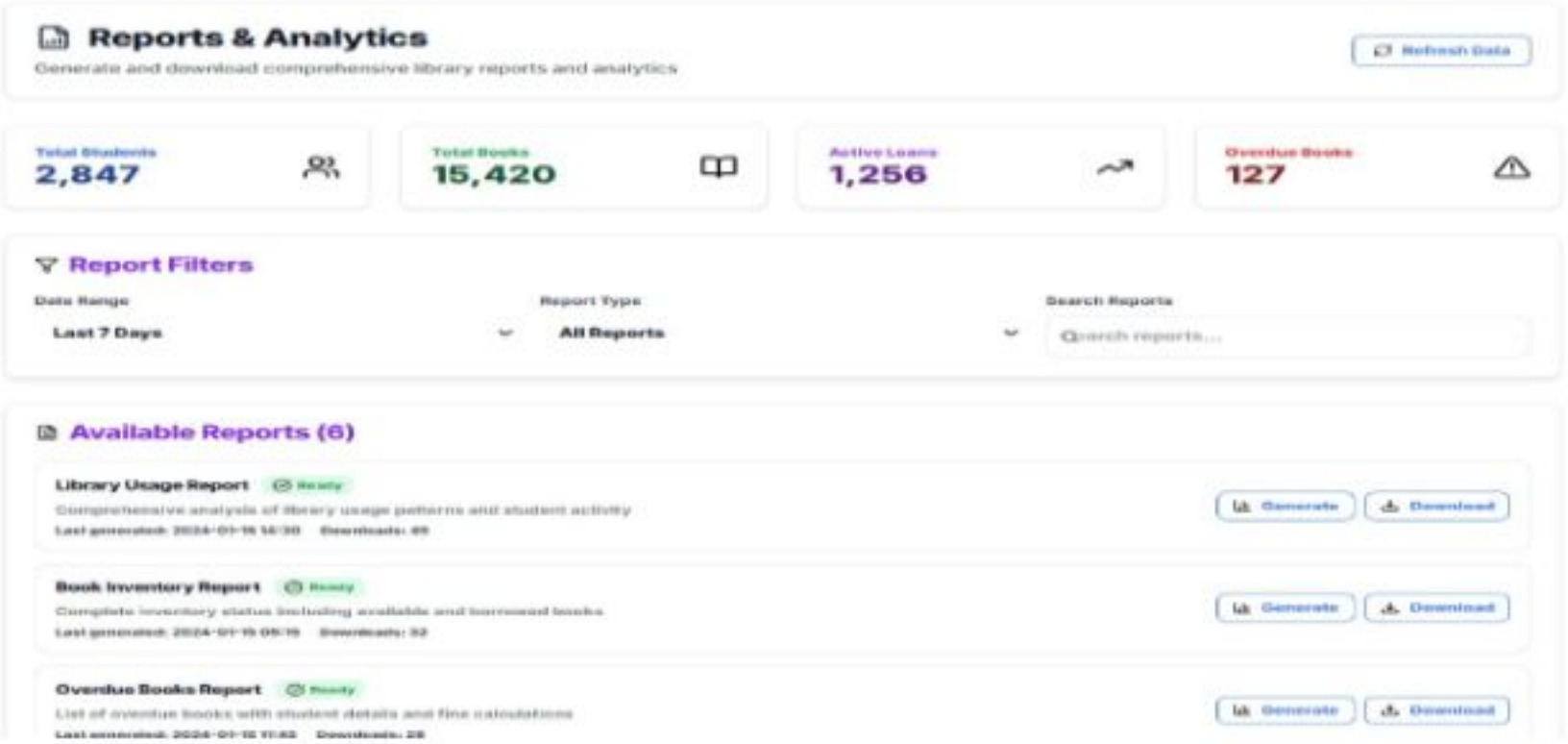
8.13 Admin Dashboard



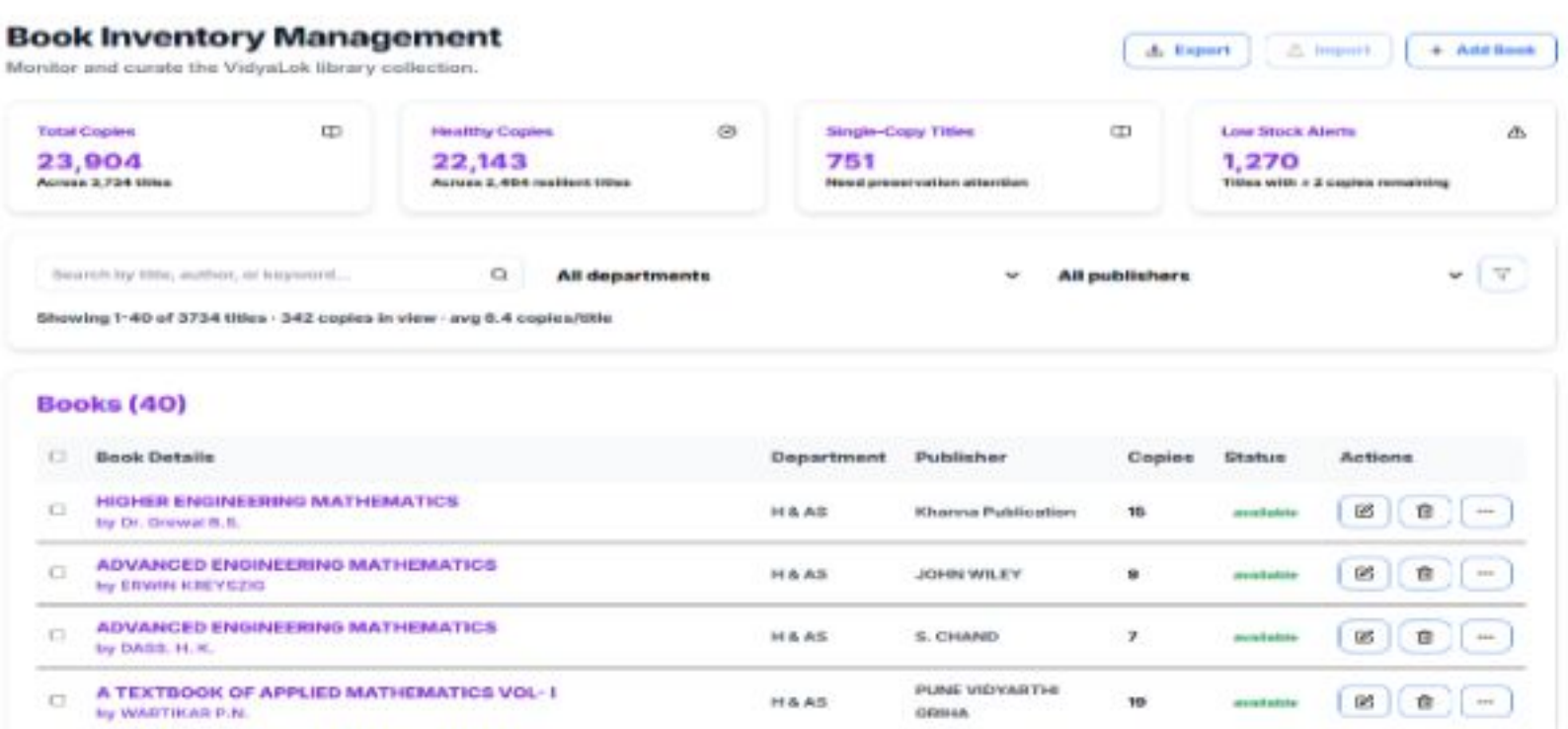
8.14 a) Analytics Dashboard



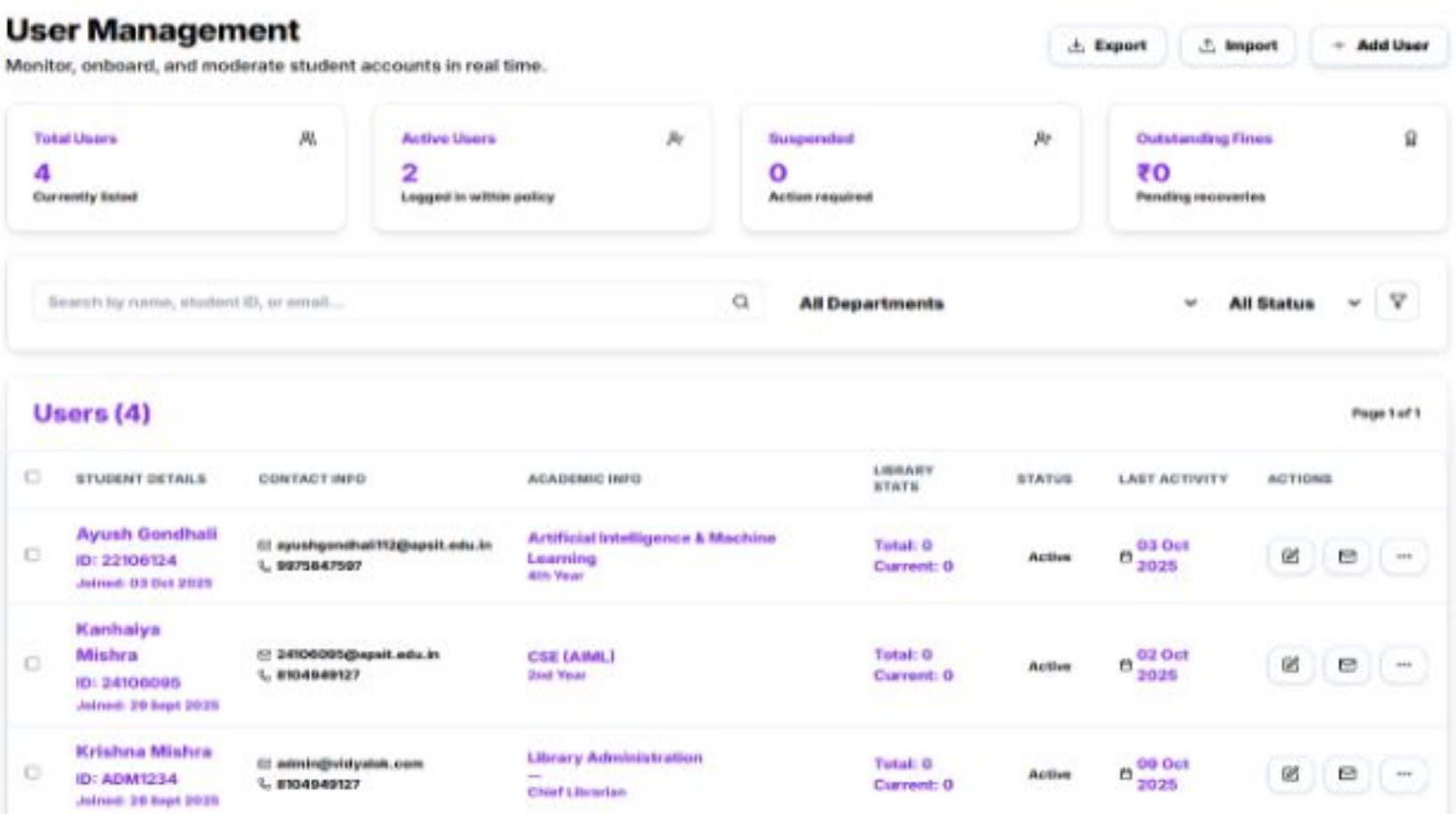
8.14 b) Analytics Dashboard



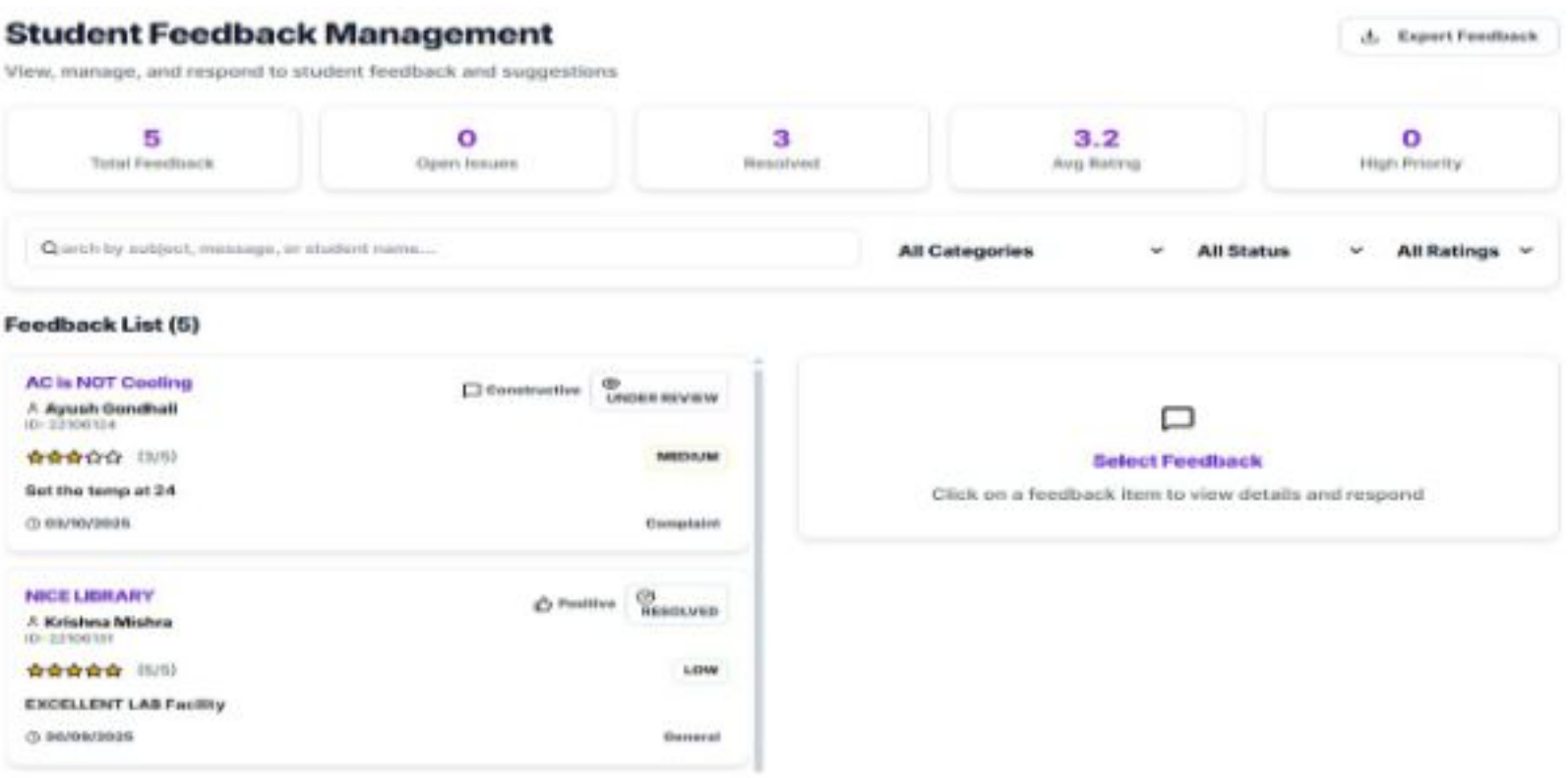
8.15 Reports and analytics for admin



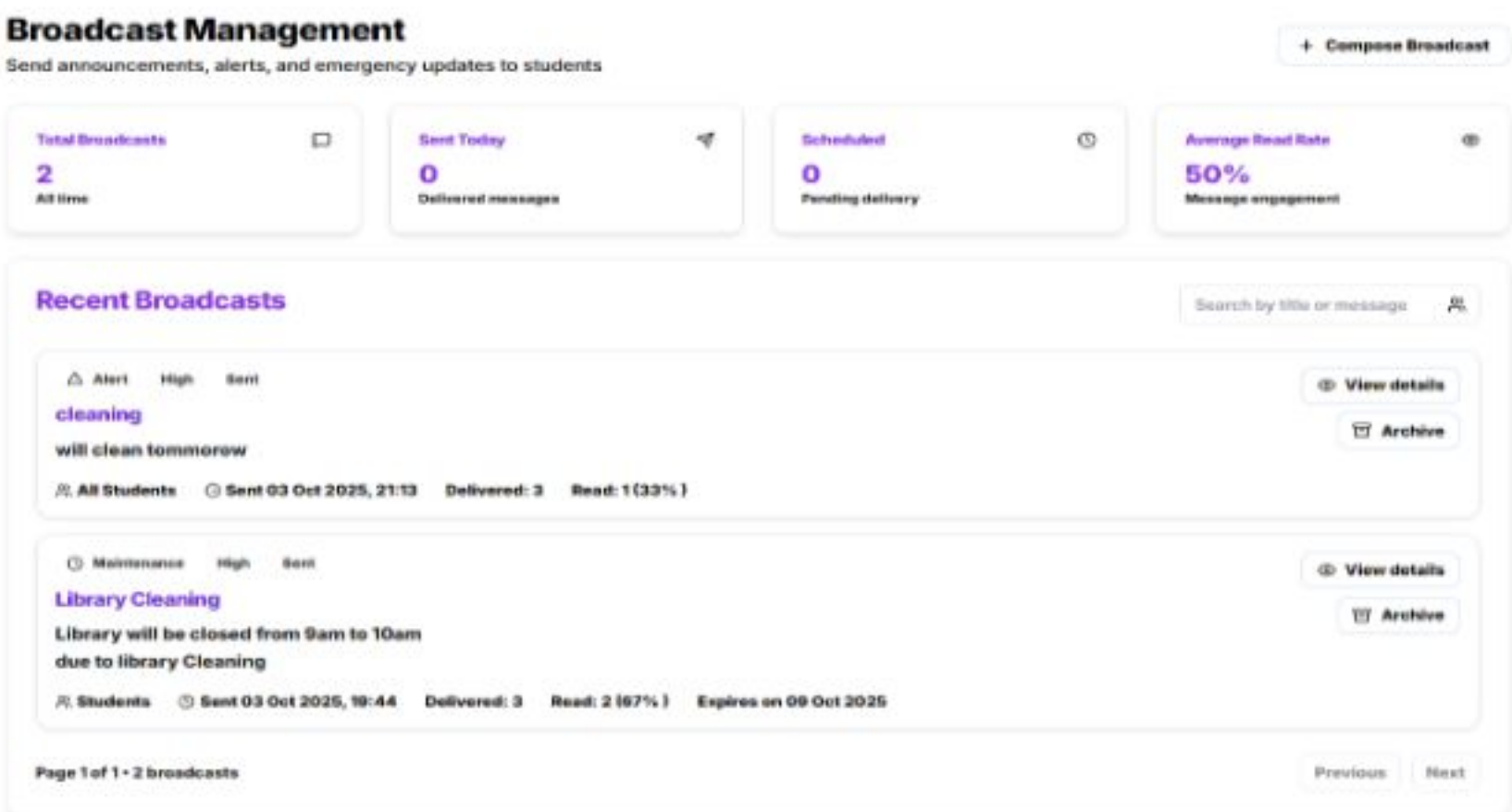
8.16 Book Inventory Management



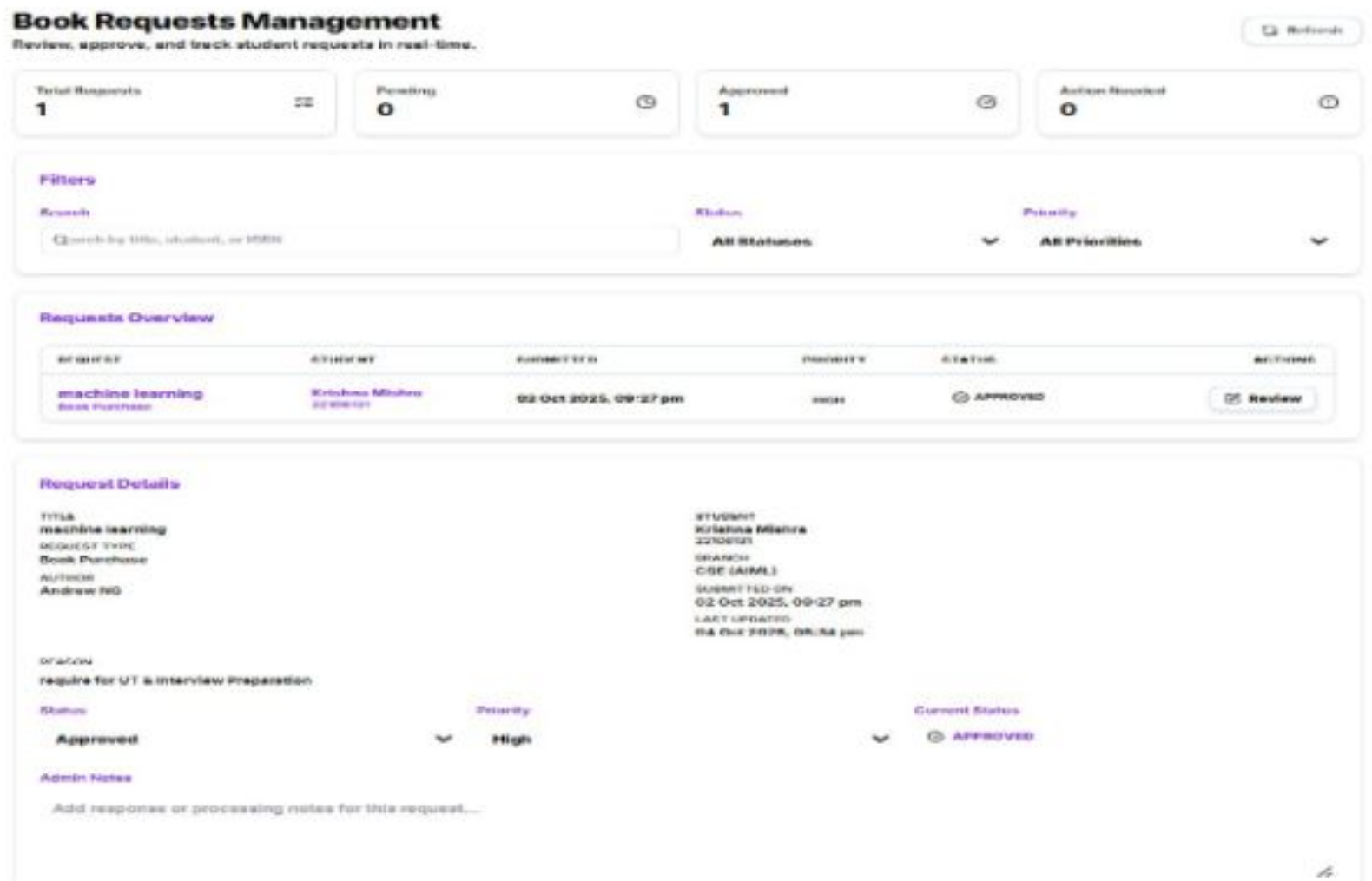
8.17 User Management to add or remove users



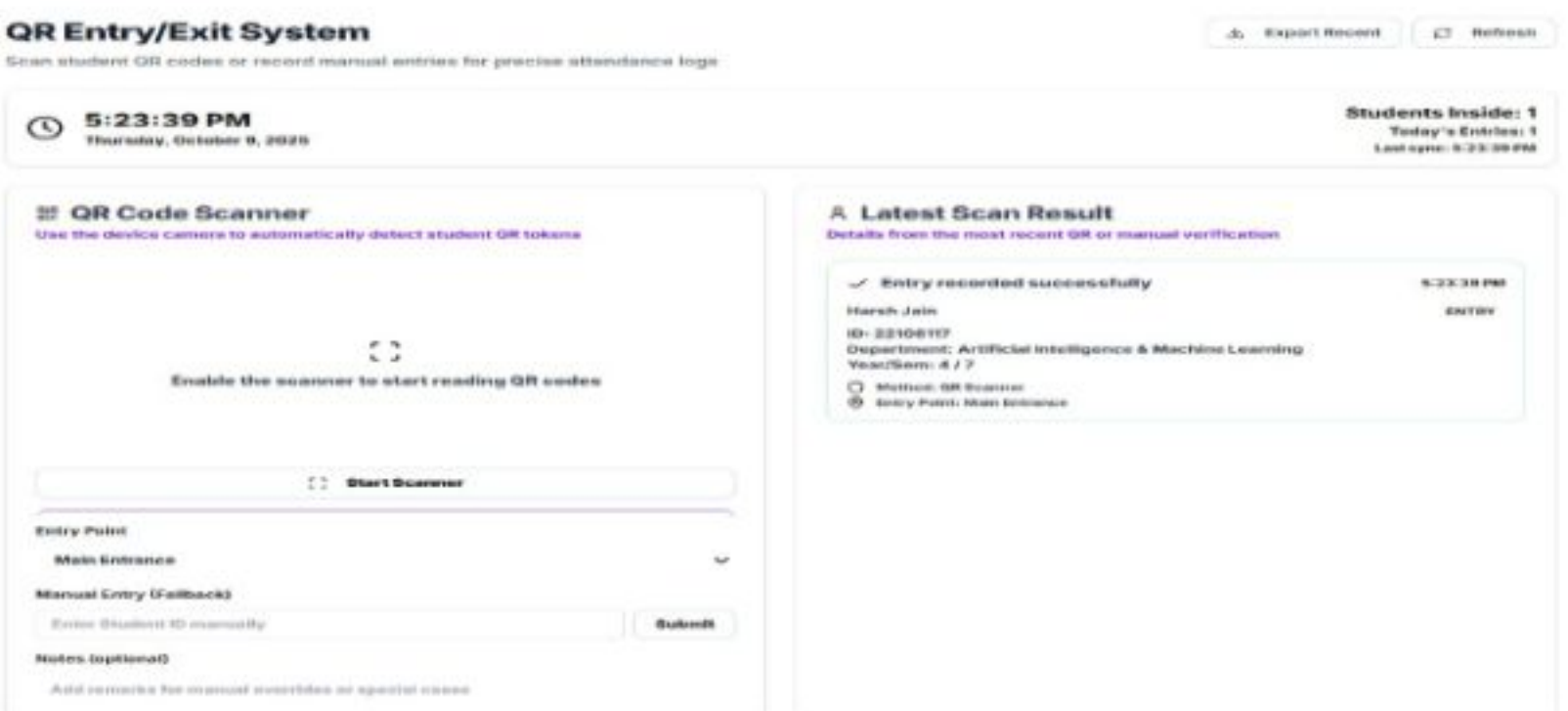
8.18 Student Feedback Management



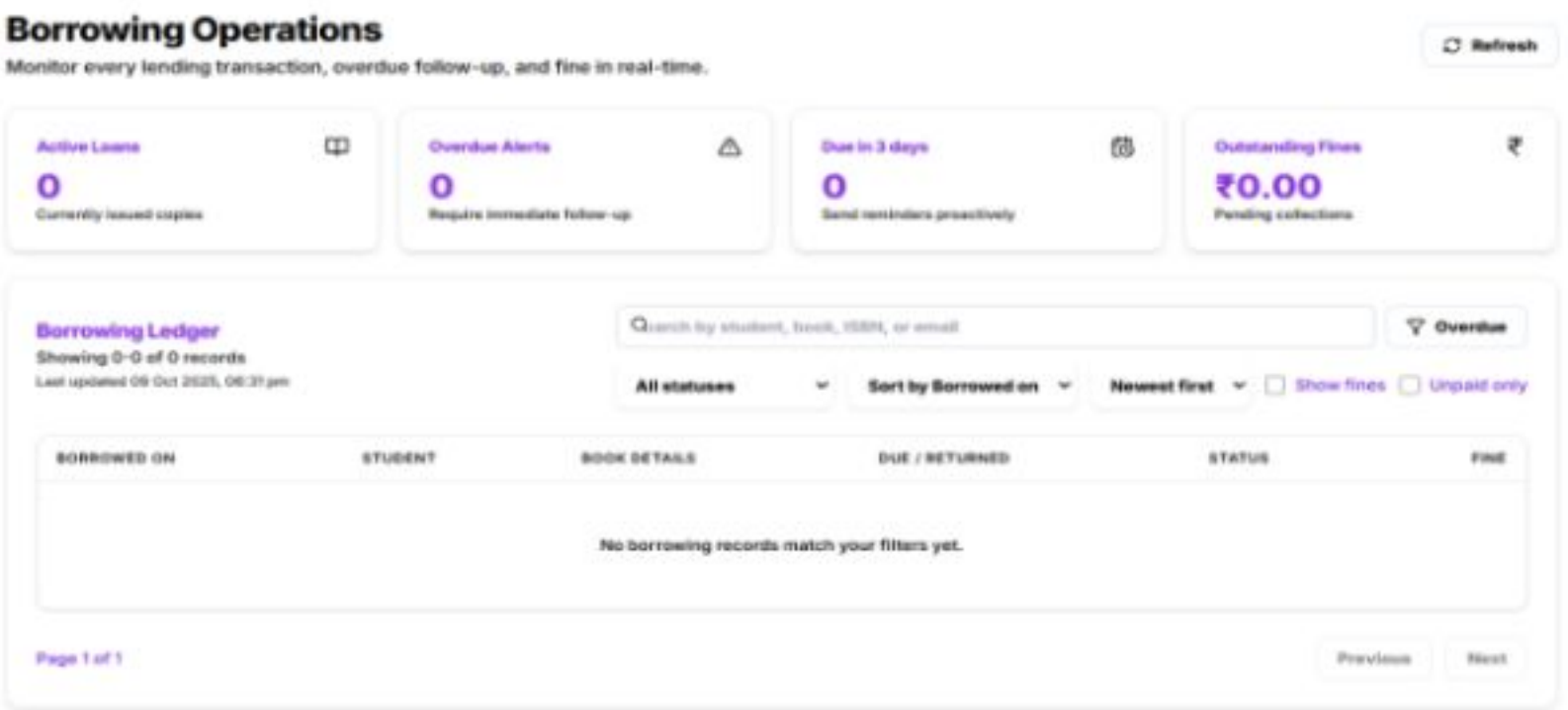
8.19 Broadcast Management



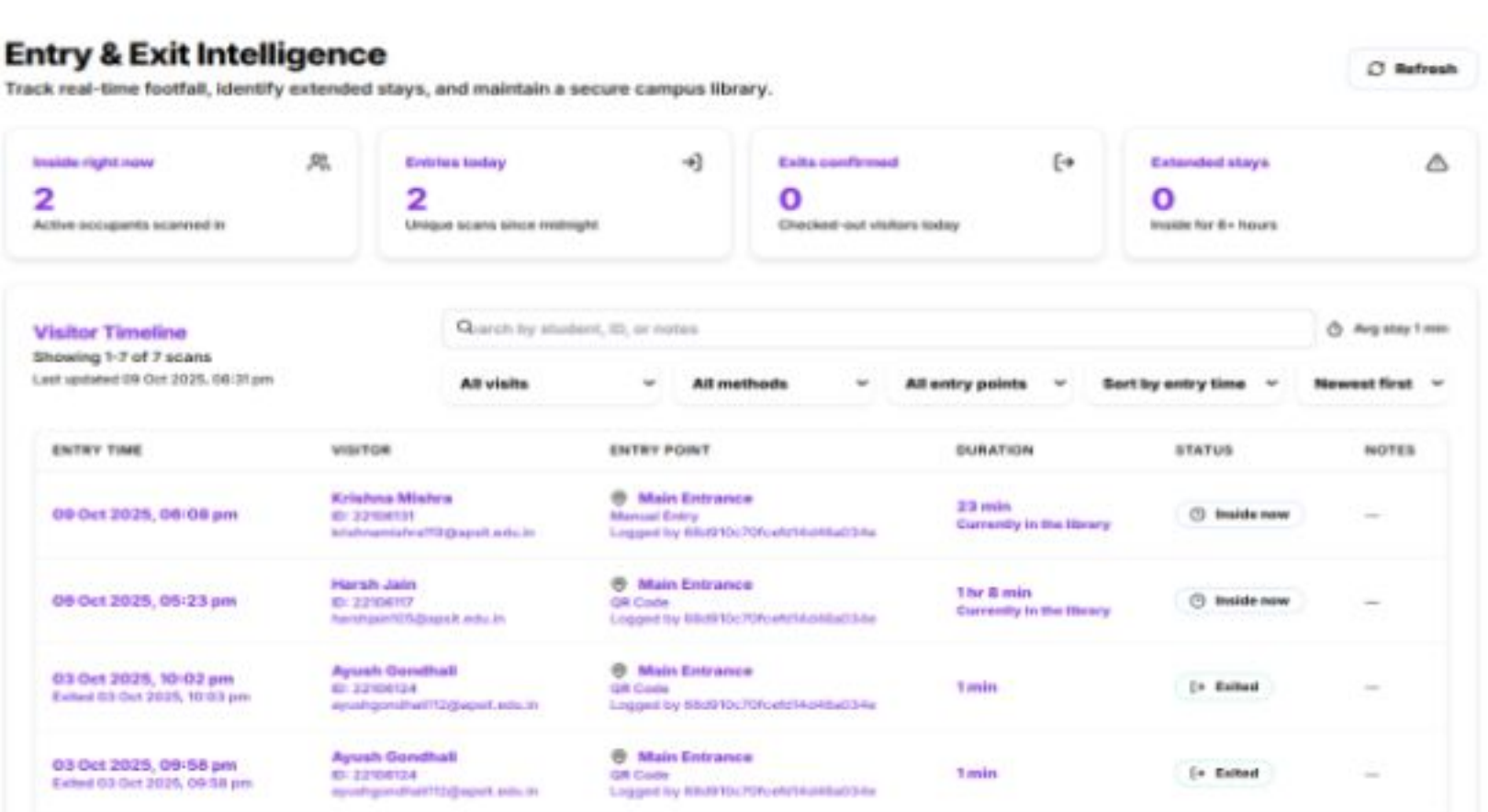
8.20 Book Request Managenent



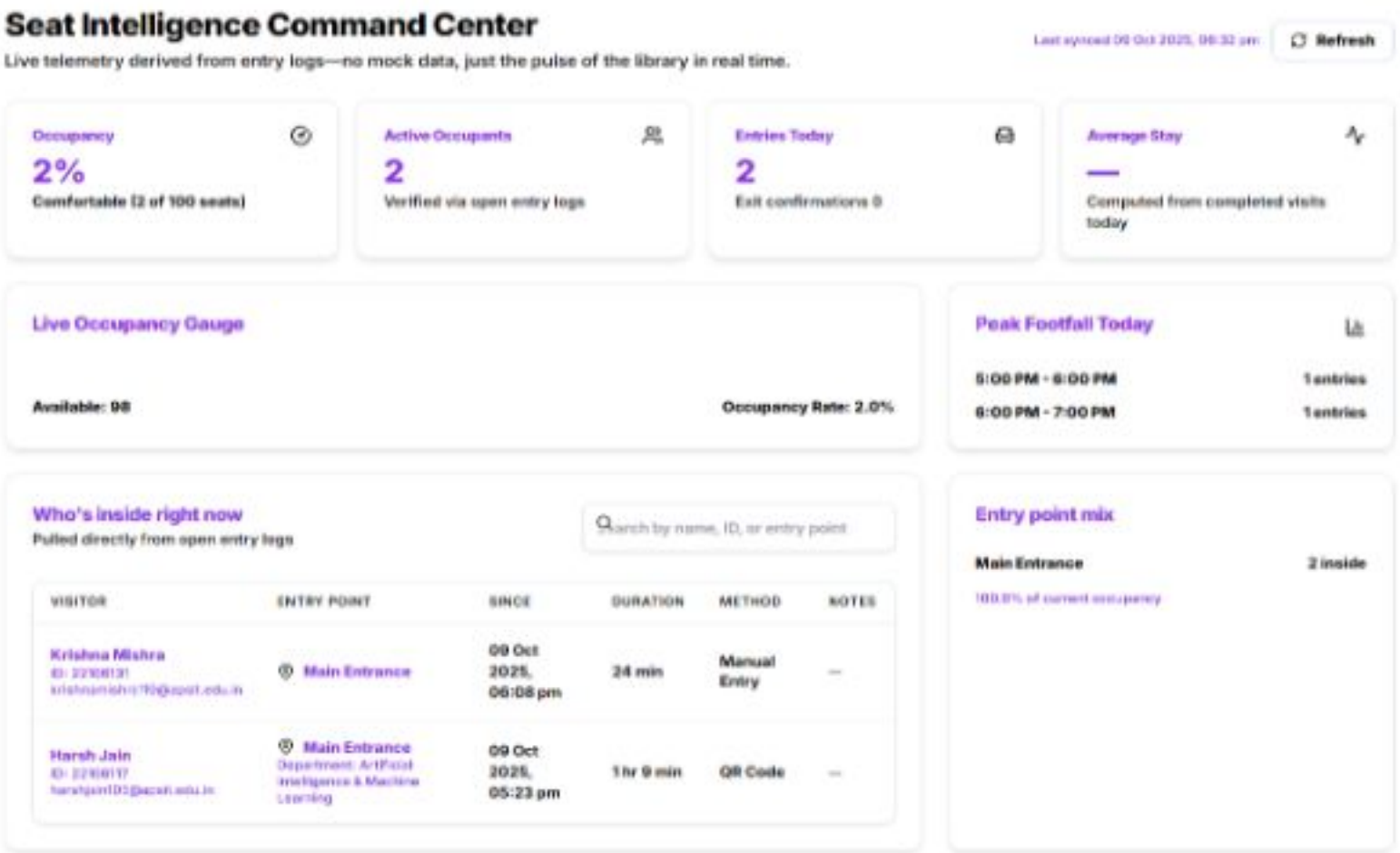
8.21 QR Entry and Exit System



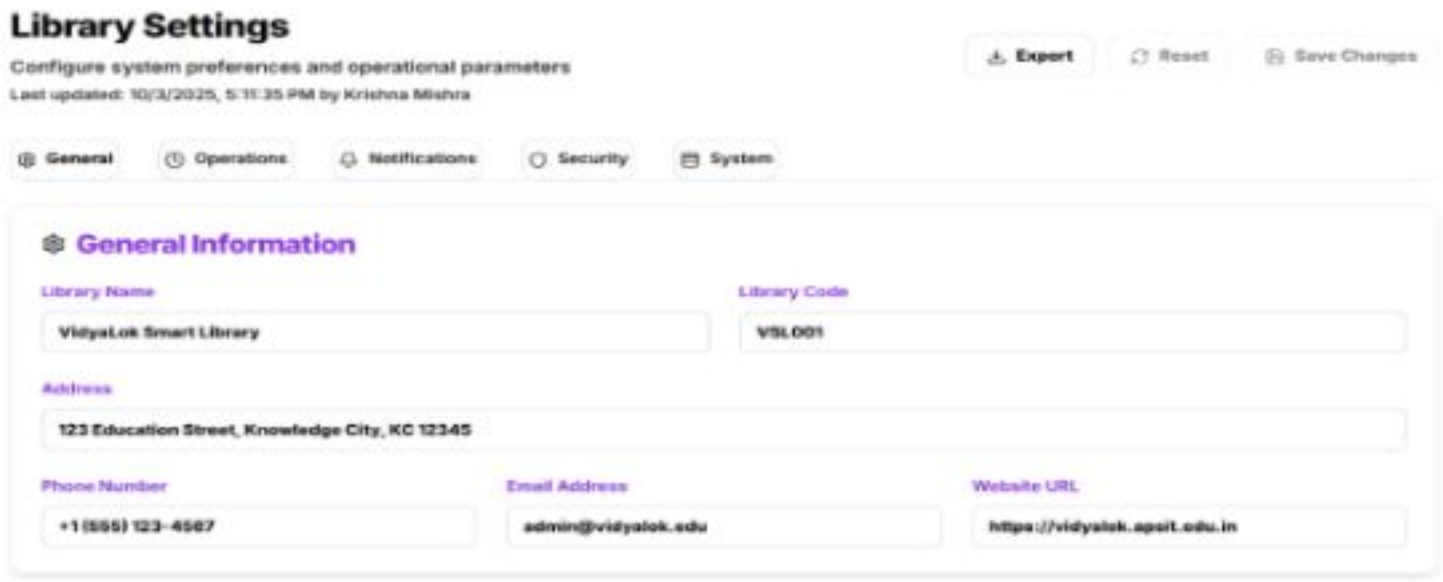
8.22 Borrowing Operations for admin



8.23 Entry and Exit Intelligence



8.24 Seat Intelligence Command Center



8.25 Library Settings

Chapter 8

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